

STAKEHOLDER 26

IT, network, and server-infrastructure teams

80 evidence-bounded questions for operational infrastructure decisions

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Asset inventory, topology and service ownership

IT-001 to IT-010

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Availability, latency, capacity and performance

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Resilience, backup, growth, energy and lifecycle planning

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Scope, safety and authority

This catalogue frames bounded questions for a future Talking Abacws service; it does not state that any capability is already deployed.

1**Proposed evidence, not installed fact**

Treat every sensor, telemetry feed, integration and analytic result as proposed until deployment, mapping, commissioning, permission and current health are independently verified. A missing signal means unknown, not normal.

2**Current authority takes precedence**

Current service owners, IT operations, cybersecurity, facilities, suppliers, change control and other competent professionals remain authoritative. The service supports decisions; it does not replace approvals or controlled procedures.

3**Historical spatial reference only**

The May 2021 floor plan may help locate historical geometry. It cannot establish current room use, access, services, hazards, closures, routes or responsibility. Current authorised records and site confirmation take precedence.

4**Protected content and control boundary**

Do not expose credentials, keys, network payloads, user content, sensitive topology or unnecessary personal data. Do not make configuration changes, restart services, grant access, diagnose users or bypass authorised IT and cybersecurity controls.

How to read each question card

Each record keeps the decision need, evidence, authority, method, uncertainty and prohibited action visible together.

Stakeholder question A natural operational question tied to a decision, not a promise that the building can already answer it.	Sensors and telemetry Physically plausible proposed evidence with location, timing, aggregation, validation, clock quality, source health, gaps, drift and privacy limits.	Authoritative supporting data Named owners, precedence, effective/observed/retrieved times, permissions, provenance, staleness and unresolved conflicts.
Analysis required Explicit lookup, observation, calculation, estimate, forecast, comparison, diagnosis support or recommendation; interval, minimum evidence, uncertainty and safe failure.	Answer boundary What the service must not infer, disclose, promise, certify, authorise or control.	Why this is useful The dispatch, escalation, validation, handback or planning action the infrastructure team can take.

Safe answer states

SUPPORTED Evidence is current and sufficient.	QUALIFIED Limitations materially affect confidence.	UNKNOWN Minimum evidence or permission is absent.	ESCALATE Authorised owner or competent professional must decide.
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Asset inventory, topology and service ownership

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IT-001

STAKEHOLDER QUESTION

What network, compute, storage and power-support assets are actually present in each authorised infrastructure room, and which asset or location records remain unverified?

Sensors and telemetry: Proposed read-only evidence: approved controller, SNMP, Redfish and virtualisation inventories; LLDP neighbour data; commissioned rack or asset tags; and metered PDU outlet state across authorised communications rooms, server rooms and racks only. Coverage/timing: Capture daily snapshots plus controller, link and power-state events; summarise one record per asset, rack position and observed power or network attachment. Quality/provenance: Validate schema and firmware versions, tag-reader or controller self-test, and asset-to-rack, port and outlet mapping. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: powered-off, isolated, unmanaged, dual-homed or mislabelled equipment can be missed; asset and rack tags require commissioning; monitor health and time alignment must be visible. Do not collect packet payloads, credentials or user content, and restrict management identifiers and location detail by role.

Analysis required: Type: lookup + observation + comparison. Decision interval: daily, after authorised moves or removals, and on demand before installation work. Minimum evidence: a current approved asset record plus one live hardware identifier, or two independent current observations tied to the same asset. Method: normalise identifiers, match rack, outlet and network attachments, separate intended from observed state, and rank unmatched or conflicting items. Uncertainty: report source age, clock quality and identifier collisions, powered-off assets, stale rack maps and unobserved rooms separately. Safe failure: return "asset or location unverified" and request controlled physical confirmation rather than asserting presence or absence.

Authoritative supporting data: Owners: Asset management, network operations, platform operations and facilities. Precedence: the approved asset register, CMDB, rack elevations and signed patch or power schedules define authorised intent; live controller, hypervisor, LLDP and PDU records show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat dynamic observations older than 24 hours, or any source with failed health checks, as stale; the May 2021 floor plan is only historical geometry. Conflicts: Expose serial, UUID, owner, rack, U-position, outlet and room conflicts; do not choose silently.

Answer boundary: Must not scan unapproved ranges, expose management addresses or sensitive topology, infer physical presence from a stale ping, or create, delete or amend asset records.

Why this is useful: The team can target physical verification, correct asset and location gaps, and plan installation or decommissioning against evidenced capacity.

IT-002

STAKEHOLDER QUESTION

Who is accountable for each critical building-facing IT service now, what cover applies, and where is the operational hand-off to facilities or suppliers?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: privacy-preserving service heartbeats, service IDs and approved dependency tags across named building-facing services and their monitored technical boundaries. Coverage/timing: Capture service state every 1-5 minutes and ownership-map refresh on catalogue or contract events; summarise service-level health and dependency presence, not individual contact behaviour. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a healthy endpoint does not prove every user journey works; unmapped services and supplier-only components may remain invisible. Do not collect private rota behaviour or expose personal contact details beyond the authorised operational role.

Analysis required: Type: lookup + comparison. Decision interval: at incident opening, before a change, after an ownership transfer and monthly. Minimum evidence: one current service ID with an accountable role, support route, hours of cover and documented hand-off point. Method: resolve aliases to the approved service, compare responsibility across technical layers, and identify missing or contradictory coverage. Uncertainty: report source age, clock quality and service aliases, expired cover, supplier-only components and stale ownership records separately. Safe failure: show the ownership gap and route to the authorised service desk or duty manager without guessing an individual.

Authoritative supporting data: Owners: Service owners, the service catalogue owner, network and platform operations, facilities and contract managers. Precedence: the approved service catalogue, accountability matrix, support model and contract demarcation define authorised intent; current duty-role records, supplier status and mapped service heartbeats show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply each rota, contract and service record validity period; mark expired cover, lapsed contracts and ownerless services as stale or incomplete. Conflicts: Expose service aliases, accountable role, support route, hours of cover and supplier-boundary conflicts; do not choose silently.

Answer boundary: Must not disclose private telephone numbers or personal rota detail, speak for a supplier, assign blame, accept contractual liability or authorise work across a demarcation point.

Why this is useful: Incidents and planned work can reach the correct accountable team sooner, with a clear hand-off package when the fault crosses organisational boundaries.

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Asset inventory, topology and service ownership

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IT-003

STAKEHOLDER QUESTION

Which connected devices are unmatched, unmanaged, retired-but-present, or otherwise outside the approved asset inventory?

Sensors and telemetry: Proposed read-only evidence: passive switch MAC learning, ARP or ND tables, NAC and RADIUS state, DHCP lease metadata and LLDP summaries from approved collection points across authorised wired and wireless infrastructure segments. Coverage/timing: Capture changes every 5-15 minutes with first-seen, last-seen and persistence summaries; summarise pseudonymous device identifiers by segment and device class. Quality/provenance: Validate units, rollover or reset semantics, interface mapping and collector version. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: randomised addresses, shared interfaces, dormant devices, unmanaged switches and monitor gaps can produce false matches or gaps. Do not inspect traffic content, derive user identity or retain identifiers longer than the approved operational purpose.

Analysis required: Type: observation + comparison. Decision interval: hourly for operational triage and weekly for controlled validation. Minimum evidence: repeated observation in at least two collection cycles plus absence from the approved inventory, or an explicit unmanaged or retired flag. Method: exclude approved transient pools, compare first and last seen times, and rank by persistence, location confidence and service exposure. Uncertainty: report source age, clock quality and randomised addresses, transient devices, monitor gaps and shared interfaces separately. Safe failure: label the item "unmatched" or "status uncertain" and request authorised validation rather than calling it malicious.

Authoritative supporting data: Owners: Cybersecurity, network operations and asset management. Precedence: the approved asset register, device-admission policy and retirement records define authorised intent; NAC decisions, controller tables and DHCP or neighbour observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the local lease, NAC and retention windows; an observation outside those windows is historical, not current. Conflicts: Expose owner, device class, approval status, retirement state and segment conflicts; do not choose silently.

Answer boundary: Must not identify a person from a device, perform unapproved fingerprinting, inspect payloads, quarantine equipment, block ports or declare a cybersecurity incident by itself.

Why this is useful: The team can separate ordinary inventory drift from persistent unmanaged equipment and route the right items through authorised validation or containment processes.

IT-004

STAKEHOLDER QUESTION

For this named service, what network and hosting path is currently evidenced, which dependencies are shared, and where does visibility stop?

Sensors and telemetry: Proposed read-only evidence: read-only routing and switching inventories, SDN or controller state, load-balancer and virtualisation mappings, and bounded synthetic reachability probes from approved test points across the service endpoints, local infrastructure path and explicitly permitted external boundary. Coverage/timing: Capture topology every 5-15 minutes, event updates where available, and low-rate probes on demand; summarise service-to-hop and service-to-platform relationships with observed alternatives. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: equal-cost paths, overlays, NAT, asymmetric routing, autoscaling and supplier or cloud internals can hide or change the path. No payloads or user sessions are inspected, and detailed topology is visible only to authorised infrastructure roles.

Analysis required: Type: lookup + observation. Decision interval: on incident demand, before maintenance and after material topology or hosting changes. Minimum evidence: a verified service ID, current endpoint mapping, at least one current topology source and a permitted test point. Method: build confirmed, probable and unobserved path segments; compare alternatives and expose the first visibility boundary. Uncertainty: report source age, clock quality and equal-cost paths, overlays, autoscaling and hidden supplier hops separately. Safe failure: return a partial path with the unobserved boundary and required owner instead of inventing hidden hops.

Authoritative supporting data: Owners: Network operations, platform operations and the service owner. Precedence: approved architecture diagrams, routing intent, deployment manifests and service dependency records define authorised intent; current controller, routing, load-balancer, orchestration and synthetic-test observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. During incidents, treat topology older than 15 minutes or probes older than the current fault window as stale. Conflicts: Expose diagram-to-live, route-to-probe, hosting-placement and ownership-boundary conflicts; do not choose silently.

Answer boundary: Must not reveal sensitive topology to unauthorised users, claim the exact path of every packet, scan beyond approved points, reroute traffic or alter hosting placement.

Why this is useful: The team can choose the right test point, identify likely shared dependencies, bound maintenance impact and prepare an evidence-based hand-off at the visibility boundary.

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Asset inventory, topology and service ownership

IT-001 to IT-010 | sheet 3 of 5

IT-005

STAKEHOLDER QUESTION

Which outlet, cable, patch-panel and switch-port records disagree with the observed connection chain?

Sensors and telemetry: Proposed read-only evidence: switch-port and LLDP state, commissioned outlet or patch-panel scans, cable-test files and approved patch records for authorised rooms and nominated outlets. Coverage/timing: retain link events and 5-15 minute port summaries; run controlled scans after patching and build one outlet-to-switch chain. Quality/provenance: use a correctly configured, in-calibration cable certifier; record both endpoints, instrument and work-record clocks, source health, untested gaps, mapping drift and signed test lineage. Limits: unmanaged endpoints, intermediate couplers, reused cables, stale labels and uncommissioned scans can break the chain. Do not expose endpoint identity or management addresses beyond authorised roles.

Analysis required: Type: comparison. Decision interval: during fault isolation, after patching and on a scheduled sample. Minimum evidence: unique identifiers at both ends, the serving port and a current link observation or cable test. Method: compare the chain, isolate the first disagreement and weight location by evidence age and label confidence. Uncertainty: report clock quality, mislabelling, undocumented patching, spare-pair use and inaccessible terminations separately. Safe failure: return "connection chain unverified" and name the next controlled test.

Authoritative supporting data: Owners: Network operations, authorised cabling contractors and facilities space-data owners. Precedence: signed patch schedules, cable certification records and current outlet or room identifiers define authorised intent; live switch and LLDP state plus controlled label scans and test results show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat mappings as stale after a move, room alteration, patching activity or failed validation; the May 2021 floor plan cannot establish current outlet use. Conflicts: Expose one-to-many labels, reused port numbers, room-name changes and observed chain conflicts; do not choose silently.

Answer boundary: Must not enable or disable ports, direct unapproved live-circuit testing, infer the user at an outlet, or amend patch records without authorised confirmation.

Why this is useful: The team can send a technician to the most likely break, reduce unnecessary room visits and correct mapping before later changes depend on the wrong port or outlet.

IT-006

STAKEHOLDER QUESTION

Where can new equipment be installed safely once rack space, power, cooling, cabling, access, weight and resilience are considered together?

Sensors and telemetry: Proposed read-only evidence: commissioned rack-position records, controlled rack scans, metered PDU load, rack-inlet temperature and environmental monitor health across candidate racks in authorised server and communications spaces. Coverage/timing: Capture power and temperature every 30-60 seconds with 5-minute aggregates, plus rack-space validation after physical changes; summarise candidate slots with associated feed, thermal and access constraints. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: blanking panels, cable obstruction, door clearance, centre-of-gravity, weight limits and undocumented reservations may not be sensed. Do not collect images unless separately authorised; use asset tags and bounded physical confirmation wherever possible.

Analysis required: Type: calculation + recommendation. Decision interval: before equipment procurement, detailed design or installation scheduling. Minimum evidence: verified contiguous U-space, equipment dimensions and weight, mapped power feeds, sustained and peak load evidence, cooling evidence and access clearance. Method: intersect hard constraints, apply approved derating and redundancy rules, and rank candidates by remaining uncertainty rather than by free U-space alone. Uncertainty: report source age, clock quality and unverified load, cooling, weight and access constraints at the proposed rack position separately. Safe failure: report no verified candidate and list the missing competent checks instead of declaring spare capacity.

Authoritative supporting data: Owners: Platform operations, network operations, facilities and competent electrical or structural professionals. Precedence: approved rack elevations, reservations, electrical design, cooling allocation, access rules and floor-loading information define authorised intent; current PDU, temperature and rack-position observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a candidate as stale after equipment moves, feed changes, cooling changes or missed sensor health; historical plans provide geometry only. Conflicts: Expose space reservation, feed capacity, cooling allocation, weight and access conflicts; do not choose silently.

Answer boundary: Must not certify electrical, structural, cooling or fire-safety suitability, reserve a rack, approve installation or instruct physical work.

Why this is useful: The team can select a feasible rack position, commission missing checks, or stop an installation before purchase and outage risk are committed.

1

Asset inventory, topology and service ownership

IT-001 to IT-010 | sheet 4 of 5

IT-007

STAKEHOLDER QUESTION

Which network, power, cooling and upstream-service dependencies support each critical asset, and which links are still assumptions?

Sensors and telemetry: Proposed read-only evidence: LLDP and switching relationships, PDU outlet mappings, UPS branch state, rack-environment sensors and approved service dependency tags across critical racks and named infrastructure assets within authorised spaces. Coverage/timing: Capture network and power state every 1-5 minutes, environmental data every 30-60 seconds, and mapping refresh after physical change; summarise asset-to-rack-to-feed-to-UPS and asset-to-switch-to-uplink dependency edges. Quality/provenance: Validate schema and firmware versions, tag-reader or controller self-test, and asset-to-rack, port and outlet mapping. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: shared circuits, dual feeds, manual transfer arrangements, hidden cooling zones and supplier services may be incompletely mapped. Restrict topology and electrical detail to authorised operational roles and collect no packet or application content.

Analysis required: Type: lookup + comparison. Decision interval: quarterly, before maintenance and during incident scoping. Minimum evidence: a verified asset or rack identity with at least one current network edge, power edge and relevant facilities dependency record. Method: join only evidence with compatible time windows, mark confirmed and assumed edges separately, and identify unproven or single-path dependencies. Uncertainty: report source age, clock quality and shared feeds, hidden upstream services, stale dependency maps and untested failover links separately. Safe failure: return an incomplete graph and the missing owner or site check rather than treating an unmapped edge as redundant.

Authoritative supporting data: Owners: Network operations, platform operations, facilities and relevant service owners. Precedence: approved single-line diagrams, rack feed schedules, network designs and service dependency records define authorised intent; current PDU, UPS, LLDP, controller and environmental observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a dependency map as stale after cabling, electrical, cooling or service-placement changes, or when any contributing monitor is unhealthy. Conflicts: Expose dual-feed status, branch mapping, upstream link, cooling-zone and service-owner conflicts; do not choose silently.

Answer boundary: Must not certify electrical isolation, expose sensitive dependency graphs broadly, operate power or cooling systems, or authorise maintenance.

Why this is useful: The team can plan maintenance blast radius, target resilience work and avoid taking a nominally redundant service down through an unrecognised shared dependency.

IT-008

STAKEHOLDER QUESTION

Which hosts, virtual machines, containers, clusters or managed platforms currently deliver each building-related digital service?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: approved hypervisor, orchestration, cluster, load-balancer and service-registry inventories with event-driven placement updates across named building-related services and authorised infrastructure platforms. Coverage/timing: Capture state every 1-5 minutes plus placement, scaling and failover events; summarise service-to-workload-to-cluster or host relationships without application content. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: autoscaling, short-lived containers, serverless functions, SaaS components and stale service tags can make placement partial or transient. Do not collect process arguments, customer data, application payloads or secrets; restrict host and tenancy detail by role.

Analysis required: Type: lookup + observation. Decision interval: during incidents, before host or cluster maintenance and after deployment changes. Minimum evidence: a verified service ID plus current workload or endpoint identifiers from at least one authoritative platform source. Method: normalise ephemeral workload IDs, group by service and environment, and distinguish active, standby, draining and historical placement. Uncertainty: report source age, clock quality and autoscaling, platform abstractions, stale service tags and supplier-only hosting layers separately. Safe failure: return the verified portion and name the unmanaged or supplier boundary rather than claiming complete placement.

Authoritative supporting data: Owners: Platform operations, the service owner and any authorised cloud or managed-platform owner. Precedence: approved deployment manifests, service registry entries and hosting responsibility records define authorised intent; current orchestrator, hypervisor, cluster and load-balancer state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat dynamic placement older than 10 minutes during incidents, or any snapshot outside its workload lifetime, as stale. Conflicts: Expose service alias, workload label, cluster, host, environment and owner conflicts; do not choose silently.

Answer boundary: Must not disclose sensitive tenancy or host detail to unauthorised users, inspect application content, migrate workloads, scale services, drain hosts or restart components.

Why this is useful: The team can identify the right platform owner, protect active workloads during maintenance and confirm whether a service actually depends on the host or cluster under investigation.

1

Asset inventory, topology and service ownership

IT-001 to IT-010 | sheet 5 of 5

IT-009

STAKEHOLDER QUESTION

What asset, topology, patching or power-map discrepancies appeared after the latest approved move, addition or removal?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: time-stamped before-and-after snapshots from approved asset, controller, LLDP, rack-position and PDU sources across the assets, racks, ports and feeds explicitly named in the authorised change scope. Coverage/timing: Capture a baseline immediately before the window, event capture during it and a post-work snapshot after monitoring stabilises; summarise expected and unexpected state differences linked to the change scope. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: missed baselines, clock offsets, concurrent work and transient discovery delay can make attribution uncertain. Do not capture secrets or configuration payloads; use identifiers, state, hashes and approved summaries only.

Analysis required: Type: comparison. Decision interval: at technical handback and again after the agreed stabilisation period. Minimum evidence: the approved scope, a usable before snapshot, an after snapshot and clocks aligned closely enough for the change window. Method: classify differences as expected, unexplained, pre-existing or time-uncertain, and link each result to its evidence. Uncertainty: report source age, clock quality and delayed telemetry, emergency work, stale baselines and incomplete handback evidence separately. Safe failure: state that attribution is indeterminate and request targeted validation when the baseline or timing is inadequate.

Authoritative supporting data: Owners: Change control, asset management, network and platform operations, facilities and the implementing owner. Precedence: the approved change record, implementation plan and expected-state list define authorised intent; time-aligned live inventories, link state, rack scans and PDU observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the authorised window and declared stabilisation period; snapshots outside it may describe unrelated work. Conflicts: Expose expected-state, observed-state, asset-owner, timestamp and concurrent-change conflicts; do not choose silently.

Answer boundary: Must not declare a person responsible, perform rollback, modify records, close the change or present temporal proximity as proof of cause.

Why this is useful: The team can correct mapping promptly, keep handback evidence focused and prevent an unexplained discrepancy from becoming the assumed baseline for later work.

IT-010

STAKEHOLDER QUESTION

Which components are supplier-managed, where is the technical demarcation, and what evidence is available on each side?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: local demarcation-interface counters, carrier or appliance health where contractually permitted, bounded synthetic probes and supplier status integrations across approved service hand-off points and the local side of supplier-managed infrastructure. Coverage/timing: Capture local state every 1-5 minutes, probes on demand and supplier status at its published cadence; summarise one service-level boundary view with local, external and unobserved portions. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: supplier portals can lag, contractual visibility can be narrow, and a healthy hand-off does not prove the remote service path. Do not collect supplier internals, credentials, payloads or information beyond the agreed operational interface.

Analysis required: Type: lookup. Decision interval: at incident triage, before supplier-affecting changes and quarterly for critical services. Minimum evidence: a current service ID, documented boundary, permitted local test and a current supplier response or status where available. Method: compare local and supplier evidence without extrapolating across the unobserved portion, and identify the next accountable owner. Uncertainty: report source age, clock quality and contract ambiguity, supplier visibility limits, stale demarcation records and hidden managed layers separately. Safe failure: report the boundary as unresolved and provide the approved escalation route rather than assigning the fault.

Authoritative supporting data: Owners: Contract owners, service owners, network or platform operations and the supplier. Precedence: the current contract, service schedule, support matrix and demarcation diagram define authorised intent; local interface tests, supplier status, incident references and authorised handback evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply contract effective dates and supplier update times; mark expired schedules, withdrawn interfaces and status older than the current incident window as stale. Conflicts: Expose demarcation location, support entitlement, service state, incident ownership and timestamp conflicts; do not choose silently.

Answer boundary: Must not speak for a supplier, inspect supplier networks, promise service credits, accept liability, authorise supplier work or disclose contract detail outside permitted roles.

Why this is useful: The team can send a concise evidence pack to the correct supplier, avoid duplicate testing and keep local restoration work separate from contractual escalation.

2

Availability, latency, capacity and performance

IT-011 to IT-020 | sheet 1 of 5

IT-011

STAKEHOLDER QUESTION

Which monitored infrastructure components are unavailable or outside their approved availability window now, and where is coverage too weak to decide?

Sensors and telemetry: Proposed read-only evidence: device and stack heartbeats, controller quorum state, critical interface state, host or cluster health and storage service checks from independent monitors across approved core, distribution, access, compute and storage components serving the building. Coverage/timing: Capture 30-60 second samples with 5-minute, hourly and daily availability summaries; summarise availability by layer, service zone and monitor path. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a failed monitor can imitate an outage; maintenance suppression, split-brain views, polling gaps and hidden supplier components limit completeness. Collect only health and state metadata, not network payloads, application content or user activity.

Analysis required: Type: observation + calculation. Decision interval: live, over 24 hours, 7 days and 30 days. Minimum evidence: at least one healthy independent monitor per component or a documented reason why only local telemetry is possible. Method: separate observed downtime, planned work, monitor blind time and unknown time; aggregate only across comparable scopes. Uncertainty: report source age, clock quality and monitor blindness, maintenance suppression, stale state and partial service coverage separately. Safe failure: show availability as indeterminate for uncovered periods rather than counting missing data as available.

Authoritative supporting data: Owners: Network operations, platform operations, storage owners and service owners. Precedence: the monitored-asset scope, service targets, approved maintenance windows and dependency maps define authorised intent; current independent monitor, controller, cluster and storage observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. During active incidents, treat state older than two collection intervals as stale; retain planned exclusions separately from raw observations. Conflicts: Expose monitor-to-device, maintenance-window, service-zone and target-definition conflicts; do not choose silently.

Answer boundary: Must not certify end-user service availability, hide unknown periods, redefine contractual exclusions, close incidents or promise compliance with a target.

Why this is useful: The team can distinguish a real outage from monitoring failure, expose blind spots and direct restoration effort to the affected layer.

IT-012

STAKEHOLDER QUESTION

Which approved paths have abnormal latency, jitter or packet loss, and when did the deviation start?

Sensors and telemetry: Proposed read-only evidence: low-rate ICMP, TCP or approved UDP synthetic probes from fixed, commissioned test points plus interface error and queue counters across defined local and external paths used for infrastructure diagnosis. Coverage/timing: Capture probes every 1-5 minutes with short higher-frequency bursts only under an authorised diagnostic procedure; summarise median, high-percentile latency, jitter, loss and route context per path. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: probe prioritisation, transient route change, Wi-Fi variability, rate limiting, clock error and Internet asymmetry can distort results. Use synthetic traffic only; do not inspect real sessions, payloads or user destinations.

Analysis required: Type: observation + comparison. Decision interval: live triage, 24-hour comparison and weekly trend assessment. Minimum evidence: two healthy test points or one test point plus corroborating interface evidence, with a usable baseline for the same path and period. Method: compare robust percentiles with time-matched baselines, identify the first deviating segment where visible, and report confidence. Uncertainty: report source age, clock quality and test-point placement, asymmetric routing, transient queues and external path changes separately. Safe failure: return "path degradation not established" and request a permitted test when baseline, route or clock evidence is insufficient.

Authoritative supporting data: Owners: Network operations and the owner of each monitored service path. Precedence: approved test-point definitions, expected route class and path-specific performance thresholds define authorised intent; current probe results, interface counters and route observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat results older than the current incident window as historical; invalidate comparisons after test-point, route or probe-method changes until a new baseline exists. Conflicts: Expose test-point mapping, route class, threshold, clock and external-boundary conflicts; do not choose silently.

Answer boundary: Must not infer what a user was doing, inspect traffic, perform unrestricted probing, guarantee the fault location or change routing, QoS or firewall policy.

Why this is useful: The team can select the right diagnostic segment, separate local from external delay and decide whether to escalate, monitor or schedule deeper authorised testing.

2

Availability, latency, capacity and performance

IT-011 to IT-020 | sheet 2 of 5

IT-013

STAKEHOLDER QUESTION

Which uplinks, trunks, port channels or virtual paths are nearing sustained capacity limits, and what resilient headroom remains?

Sensors and telemetry: Proposed read-only evidence: interface byte and packet counters, queue drops, LAG member utilisation, virtual-switch counters and transceiver state across approved core, distribution, access and relevant virtual network links. Coverage/timing: Capture 30-60 second polling with 5-minute aggregates and event capture for member loss; summarise peak, 95th and 99th percentile utilisation, directional imbalance and resilient-capacity headroom. Quality/provenance: Validate units, rollover or reset semantics, interface mapping and collector version. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: counter wraps, polling gaps and sub-minute microbursts can be missed; logical capacity may differ from line rate because of overhead, QoS and failed members. No payload, endpoint content or user-level flow detail is required.

Analysis required: Type: calculation + estimate. Decision interval: hourly triage, weekly capacity assessment and before major events or changes. Minimum evidence: a verified link identity, current effective capacity, at least seven representative days or a justified shorter window, and known resilience rules. Method: calculate directional percentiles, repeat-duration above thresholds and headroom after the largest credible member loss. Uncertainty: report source age, clock quality and counter resets, burst aggregation, effective-capacity rules and failed-member states separately. Safe failure: state that headroom is unknown or provisional when effective capacity, workload representativeness or burst evidence is inadequate.

Authoritative supporting data: Owners: Network operations and capacity owners. Precedence: approved link design, LAG membership, reserved capacity, QoS policy and resilience requirement define authorised intent; current interface, queue, member-state and virtual-switch observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat membership or capacity records as stale immediately after link, optics or configuration change; flag monitor gaps longer than two intervals. Conflicts: Expose configured speed, effective capacity, member count, reservation and counter-source conflicts; do not choose silently.

Answer boundary: Must not reconfigure trunks, change QoS, promise throughput, infer application content or expose sensitive traffic distribution outside authorised roles.

Why this is useful: The team can schedule upgrades before recurring congestion, rebalance planned demand and distinguish acceptable peaks from links that lose resilience under normal load.

IT-014

STAKEHOLDER QUESTION

Which access or server-facing ports show persistent errors, discards, optical degradation, speed mismatch or link instability that merits a physical check?

Sensors and telemetry: Proposed read-only evidence: port counters, link-up or down events, negotiated speed and duplex, transceiver digital diagnostics and approved switch-stack health across authorised access, server and uplink ports. Coverage/timing: Capture 30-60 second counters plus immediate link events and daily persistence summaries; summarise error rates and flapping by port, switch, optics and connected asset class. Quality/provenance: Validate units, rollover or reset semantics, interface mapping and collector version. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: sleeping endpoints, intentional cable removal, counter resets, unsupported optics and dirty fibre can resemble faults; endpoint drivers may be unobserved. Restrict port-to-asset mappings by role and collect no packet content.

Analysis required: Type: observation + comparison. Decision interval: live triage and daily exception ranking. Minimum evidence: at least two consecutive intervals or repeated events plus a verified port identity and counter reset context. Method: normalise errors by traffic volume, separate physical indicators from congestion drops, and rank ports by recurrence and service importance. Uncertainty: report source age, clock quality and counter semantics, endpoint resets, optics-telemetry accuracy and intermittent faults separately. Safe failure: label the port "needs controlled validation" when counters cannot distinguish cable, optics, switch or endpoint causes.

Authoritative supporting data: Owners: Network operations and the relevant platform or endpoint asset owner. Precedence: approved port purpose, configured speed, optics type and maintenance state define authorised intent; current switch counters, link events, transceiver telemetry and controlled test results show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Reset baselines after authorised port or optics replacement and mark counters with unknown reset time as unsuitable for rate comparison. Conflicts: Expose port purpose, endpoint mapping, configured speed, optics identity and maintenance-state conflicts; do not choose silently.

Answer boundary: Must not disable ports, reset interfaces, identify or diagnose a user, direct unsafe cable work or declare hardware failure without authorised testing.

Why this is useful: The team can prioritise the ports most likely to benefit from a cable, optics or endpoint check and avoid replacing equipment based on a single misleading counter.

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Availability, latency, capacity and performance

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IT-015

STAKEHOLDER QUESTION

Are DNS, DHCP, time, authentication or other shared services delaying connection setup, and at which stage does the delay appear?

Sensors and telemetry: Proposed read-only evidence: dedicated synthetic DNS queries, DHCP test transactions, NTP offset checks, approved authentication test identities and service health counters across commissioned test segments and shared infrastructure endpoints. Coverage/timing: Capture transactions every 1-5 minutes with stage timings and 5-minute aggregates; summarise success rate and response time by service stage and test point. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: test identities can differ from real clients, caches can mask faults, rate limits can affect probes and a successful transaction does not prove every policy path. Use dedicated synthetic identities; do not capture real usernames, credentials, queries, leases or user content.

Analysis required: Type: diagnosis support. Decision interval: live incident triage and daily service assurance. Minimum evidence: a successful control transaction plus at least one failing or slow transaction from a healthy test point, or corroborating server-side health evidence. Method: compare each stage with its baseline, align clocks and identify whether delay is local, shared-service or external. Uncertainty: report source age, clock quality and transaction-stage coverage, cache effects, timeout policy and shared-service visibility separately. Safe failure: report only the failing or uncertain stage and escalate to its owner rather than diagnosing a user or declaring root cause.

Authoritative supporting data: Owners: Owners of DNS, DHCP, time, identity and network services. Precedence: approved service endpoints, test identities, expected transaction sequence and service thresholds define authorised intent; current synthetic results, server health, queue or replication state and permitted dependency observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate a baseline after endpoint, policy or test-identity changes until enough new observations exist; use each service owner's freshness rule. Conflicts: Expose endpoint, service-role, threshold, test-identity and stage-timing conflicts; do not choose silently.

Answer boundary: Must not use real user credentials, expose directory information, alter leases or records, restart shared services or promise that all client types are unaffected.

Why this is useful: The team can avoid chasing access switches when the delay sits in a shared service and can send the responsible owner a time-aligned transaction trace without user data.

IT-016

STAKEHOLDER QUESTION

Which hosts, clusters or hypervisors face sustained CPU, memory, storage-I/O or network-I/O pressure, and which services are exposed?

Sensors and telemetry: Proposed read-only evidence: host and cluster CPU utilisation, run queue, memory pressure, swap, datastore latency, network-I/O and scheduler contention metrics across authorised physical hosts, hypervisors and clusters. Coverage/timing: Capture 30-60 second samples with 5-minute and hourly aggregates plus host-state events; summarise resource pressure by host or cluster and approved service class. Quality/provenance: Validate units, rollover or reset semantics, interface mapping and collector version. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: short bursts, noisy-neighbour effects, ballooning, guest-agent gaps and hidden managed-platform metrics can make workload attribution incomplete. Do not collect process arguments, file names, application payloads or individual user activity.

Analysis required: Type: observation + calculation. Decision interval: live triage, hourly ranking and weekly capacity assessment. Minimum evidence: current placement, at least three aligned resource metrics and enough continuous data to distinguish a burst from sustained pressure. Method: compare against host-specific baselines and approved limits, identify co-occurring bottlenecks and distinguish capacity from scheduling or storage effects. Uncertainty: report source age, clock quality and virtualisation attribution, burst masking, workload migration and metric saturation separately. Safe failure: return "pressure observed; workload impact unverified" when service-level evidence is absent.

Authoritative supporting data: Owners: Platform operations, virtualisation owners and relevant service owners. Precedence: approved host or cluster membership, reservations, limits, maintenance state and service placement records define authorised intent; current host, hypervisor, orchestrator and storage observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat host placement and cluster membership as stale after migration, failover or maintenance events; mark gaps longer than two intervals. Conflicts: Expose host identity, workload placement, reservation, metric definition and maintenance-state conflicts; do not choose silently.

Answer boundary: Must not expose tenant detail beyond authorised roles, kill processes, migrate workloads, change reservations or diagnose application code.

Why this is useful: The team can choose whether to inspect a host, cluster, storage path or workload owner first and can schedule capacity work before sustained contention becomes an outage.

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Availability, latency, capacity and performance

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IT-017

STAKEHOLDER QUESTION

Which storage pools, volumes, datastores or backup repositories are close to practical capacity or showing sustained latency, queue or error degradation?

Sensors and telemetry: Proposed read-only evidence: allocated and physical capacity, thin-provisioning commitments, IOPS, throughput, latency, queue depth, replication state and storage error counters across authorised production, infrastructure and backup storage systems. Coverage/timing: Capture 1-5 minute samples with hourly and daily capacity summaries plus fault events; summarise usable headroom, growth rate and performance by pool or service class. Quality/provenance: Validate units, rollover or reset semantics, interface mapping and collector version. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: deduplication, compression, snapshots, delayed reclamation, cache effects and vendor-specific counters can make apparent and usable capacity differ. Do not inspect file names, object names, database content, backup content or tenant data.

Analysis required: Type: observation + estimate. Decision interval: daily for capacity, live for faults and weekly for trend decisions. Minimum evidence: current usable-capacity definition, at least 30 representative days for forecasting where available, and aligned latency, queue and error evidence for performance claims. Method: calculate conservative time-to-threshold ranges, separate transient cache effects from sustained pressure and test sensitivity to growth assumptions. Uncertainty: report source age, clock quality and thin provisioning, cache effects, replication state and repository reservations separately. Safe failure: state that exhaustion date or fault location is indeterminate when reclamation, compression or workload evidence is missing.

Authoritative supporting data: Owners: Storage owners, platform operations, backup owners and service owners. Precedence: approved pool design, reservation, replication policy, alert thresholds and service allocation define authorised intent; current array, software-defined storage, hypervisor and backup-platform observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat allocation or reservation data as stale after provisioning, snapshot, replication or policy change; retain metric definitions with the evidence. Conflicts: Expose logical-versus-physical capacity, reservation, replication, ownership and metric-definition conflicts; do not choose silently.

Answer boundary: Must not delete data, expand volumes, alter replication, expose stored content or promise recoverability or performance.

Why this is useful: The team can plan expansion, cleanup or workload movement through authorised processes and can distinguish a capacity problem from a latency or replication problem.

IT-018

STAKEHOLDER QUESTION

Which recurring time windows show infrastructure degradation, and which shared conditions repeatedly coincide without proving cause?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: aggregated network, compute, storage, shared-service, power and cooling metrics together with approved maintenance and event calendars across the infrastructure supporting the affected service or zone. Coverage/timing: Capture 1-5 minute operational metrics retained as hourly and time-of-week summaries over 30-90 days; summarise recurrence windows and co-occurring infrastructure conditions. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: seasonal demand, monitor changes, missing intervals and unrelated scheduled activity can produce spurious patterns. Use aggregate service and infrastructure demand only; do not infer attendance, productivity or individual behaviour.

Analysis required: Type: comparison + association. Decision interval: monthly and after repeated degradation reports. Minimum evidence: at least three comparable recurrences or a sufficiently long baseline with documented coverage and clock quality. Method: use robust baselines, cluster similar windows, test whether associations persist across occurrences and report alternative explanations. Uncertainty: report source age, clock quality and seasonality, coincident demand, maintenance schedules and changing baselines separately. Safe failure: report a pattern as unconfirmed when recurrence count, coverage or comparability is inadequate.

Authoritative supporting data: Owners: Network, platform, storage and facilities operations plus the affected service owner. Precedence: approved metric definitions, maintenance windows, service calendar and known exceptional events define authorised intent; time-aligned operational telemetry and incident or event references show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Exclude periods with changed monitors or incomparable configurations unless separately modelled; retain the observation range and missing-data rate. Conflicts: Expose metric-definition, timezone, event-window, service-scope and maintenance-classification conflicts; do not choose silently.

Answer boundary: Must not infer people's behaviour, label a team or application as the cause, inspect user content, or automatically alter capacity or schedules.

Why this is useful: The team can schedule targeted measurement or controlled tests in the recurring window instead of attributing the slowdown to the most visible coincident event.

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Availability, latency, capacity and performance

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IT-019

STAKEHOLDER QUESTION

Does each critical service meet its approved availability, response-time and capacity-headroom targets over the correct scope and measurement window?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: approved service heartbeats, synthetic transactions, dependency health, interface or platform capacity metrics and monitor-coverage state across the exact service endpoints and infrastructure components named in each target. Coverage/timing: Capture 1-5 minute observations with daily and monthly target-window aggregates; summarise numerator, denominator, exclusions, unknown time and confidence for each target. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: target scope, maintenance treatment, monitor blind time and client-path diversity can materially change the result. Use synthetic and infrastructure evidence only; do not collect user transactions or content.

Analysis required: Type: calculation + comparison. Decision interval: weekly operational tracking and at the target reporting interval. Minimum evidence: a current target definition, matching monitored scope, documented exclusions and enough valid observations for the stated window. Method: recompute from raw states, separate planned, unplanned and unknown intervals, and provide sensitivity where exclusions are disputed. Uncertainty: report source age, clock quality and target scope, excluded periods, sampling bias and incompatible measurement definitions separately. Safe failure: state that target attainment cannot be established when scope or coverage is not valid.

Authoritative supporting data: Owners: Service owners, service-level governance and the responsible infrastructure teams. Precedence: the approved target definition, scope, measurement method, exclusions and effective dates define authorised intent; current monitoring, synthetic-test and dependency evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Do not apply a superseded target to a current period; mark any interval with insufficient monitor coverage as unknown rather than compliant. Conflicts: Expose scope, formula, exclusion, endpoint, timezone and target-version conflicts; do not choose silently.

Answer boundary: Must not rewrite targets, certify contractual compliance, suppress unknown time, promise future performance or approve service acceptance.

Why this is useful: The team can separate a service miss from weak measurement and give the accountable owner a defensible target-versus-evidence package.

IT-020

STAKEHOLDER QUESTION

Did infrastructure performance materially change after a deployment, configuration change, maintenance activity or unusual event?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: time-aligned before-and-after synthetic tests, interface, host, storage and shared-service metrics, plus monitor-health and route or placement context across the services and dependencies within the approved event or change scope. Coverage/timing: Capture continuous 1-5 minute data spanning a defined baseline, event window and stabilisation period; summarise matched performance indicators and change points by service and dependency. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: concurrent changes, route shifts, autoscaling, demand differences and short baselines can confound comparison. Do not inspect application payloads or individual sessions; use approved aggregate and synthetic measures.

Analysis required: Type: comparison. Decision interval: at technical handback, after stabilisation and when degradation is reported. Minimum evidence: a valid pre-event baseline, aligned post-event data, stable monitor definitions and at least one corroborating metric for a material-change claim. Method: compare robust distributions rather than single averages, control for known demand or route differences and report effect range with alternatives. Uncertainty: report source age, clock quality and baseline drift, overlapping changes, unusual demand and delayed metric availability separately. Safe failure: state "performance change not attributable" when timing or comparability is inadequate.

Authoritative supporting data: Owners: Change control, infrastructure owners and the affected service owner. Precedence: the authorised event record, expected impact, acceptance criteria and baseline definition define authorised intent; current time-series telemetry, synthetic tests and topology or placement observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use only comparable periods and reset baselines after architecture or metric-definition changes; flag clock or coverage gaps. Conflicts: Expose event timing, scope, baseline, demand condition, topology and metric-definition conflicts; do not choose silently.

Answer boundary: Must not prove causation from timing alone, perform rollback, alter configuration, diagnose application code or close the change or incident.

Why this is useful: The team can decide whether to continue observation, test a specific dependency, invoke rollback through control procedures or investigate an unrelated condition.

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Wired, wireless and building-network coverage

IT-021 to IT-030 | sheet 1 of 5

IT-021

STAKEHOLDER QUESTION

Which room data outlets are correctly mapped, enabled and electrically certified, and which required locations still lack a verified wired connection?

Sensors and telemetry: Proposed read-only evidence: commissioned outlet and patch-panel identifiers, switch-port state, approved cable-certifier results and bounded technician test-client observations across nominated rooms, desks, teaching spaces, plant areas and service points where wired connectivity is required. Coverage/timing: Capture link state every 5-15 minutes and controlled certification or test results after installation, alteration or fault work; summarise one outlet-to-patch-to-port chain with enablement and test status. Quality/provenance: Use a cable certifier with a current calibration certificate, correct test limit and recorded outlet and patch endpoints; verify instrument clock alignment to the work record. Continuous uptime is not applicable: expose untested gaps and later mapping drift, and retain the signed test file and provenance. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: an unused outlet may show no link, labels may be stale, and switch state cannot prove cable performance or physical accessibility. Do not associate an outlet with an individual or inspect test traffic beyond the synthetic connectivity check.

Analysis required: Type: lookup + observation. Decision interval: on demand before occupation or equipment installation, after alterations and during fault triage. Minimum evidence: a current room requirement, uniquely mapped outlet chain and either a valid certification result or controlled live test. Method: separate installed, enabled, tested, failed and unverified states; do not treat link absence as cable failure. Uncertainty: report source age, clock quality and outlet mislabelling, patch changes, inaccessible rooms and expired certification separately. Safe failure: return the location as unverified and specify the next permitted physical or electrical test.

Authoritative supporting data: Owners: Network operations, authorised cabling contractors, facilities space-data owners and the requesting service owner. Precedence: current outlet schedules, cable certification, approved room identifiers and service requirements define authorised intent; live switch state, controlled port tests and commissioned label observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate a mapping after room alteration, repatching or failed certification; the May 2021 floor plan is historical reference only. Conflicts: Expose room name, outlet label, patch position, switch port, enablement and certification conflicts; do not choose silently.

Answer boundary: Must not enable ports, certify cabling, infer who uses an outlet, authorise access to rooms or instruct unapproved work on live equipment.

Why this is useful: The team can dispatch the correct cabling or network action, avoid promising service from an untested outlet and identify where a new connection is genuinely required.

IT-022

STAKEHOLDER QUESTION

Where does managed Wi-Fi fall below the agreed RF and service threshold, and which spaces are too stale or unmeasured to assess?

Sensors and telemetry: Proposed read-only evidence: calibrated survey clients, AP radio telemetry, channel utilisation, noise, retries and bounded synthetic service tests across authorised teaching, meeting, public, circulation and operational spaces included in the wireless design. Coverage/timing: Capture AP aggregates every 5 minutes and controlled walk surveys after material change or at the approved periodic interval; summarise RSSI, SNR, channel use, retry rate and synthetic success by survey cell or route. Quality/provenance: Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: walls, doors, furniture, crowds, device diversity, external interference and survey-client calibration can shift results; interpolation beyond measured cells is uncertain. Use dedicated survey devices and aggregate AP data; do not track client identities, movements, content or destinations.

Analysis required: Type: observation + comparison. Decision interval: after reported problems, after relevant changes and at the approved survey cycle. Minimum evidence: a calibrated survey device, current zone map, verified AP state and measurements dense enough for the claimed space. Method: compare measured values with the design threshold, preserve raw variability, and distinguish RF weakness from authentication or upstream-service failure. Uncertainty: report source age, clock quality and survey age, client differences, temporary interference and post-survey building changes separately. Safe failure: mark unmeasured or stale cells unknown rather than colouring them as covered.

Authoritative supporting data: Owners: Wireless network operations and facilities space-data owners. Precedence: the approved wireless design, target thresholds, current zone geometry and intended AP placement define authorised intent; commissioned survey results and current controller or AP observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a survey as stale after partition, furniture, AP, power or room-use change and according to the wireless owner's maximum survey age; May 2021 geometry cannot establish current conditions. Conflicts: Expose zone boundary, AP placement, threshold, survey date and controller-to-survey conflicts; do not choose silently.

Answer boundary: Must not identify or track users, infer room occupancy, guarantee performance for every device, change radio settings or certify full building coverage.

Why this is useful: The team can choose targeted survey routes, AP or channel checks and building-change follow-up without relying on anecdotal reports or an outdated heat map.

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Wired, wireless and building-network coverage

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IT-023

STAKEHOLDER QUESTION

Can a managed test client roam along the operational routes that matter without abnormal interruption, reauthentication delay or session loss?

Sensors and telemetry: Proposed read-only evidence: dedicated test-client association events, AP transitions, authentication-stage timing, packet-loss and latency probes across defined routes such as entrances to teaching spaces, service corridors and incident-command locations where roaming is operationally required.

Coverage/timing: Capture controlled route tests quarterly, after WLAN changes and when a repeatable roaming problem is reported; summarise handover time, loss burst, reauthentication time and AP sequence per route segment. **Quality/provenance:** Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. **Limits:** a single test device cannot represent every chipset; walking speed, doors, crowds, power save and band steering can change results. Use only commissioned test clients; do not reconstruct real users' movements or sessions.

Analysis required: Type: observation + comparison. Decision interval: after changes, during focused fault investigation and at the agreed validation interval. Minimum evidence: a current route, healthy APs, a commissioned test profile and at least two repeat traversals in each direction. Method: align association and probe events, compare repeatability, locate the transition zone and report device-specific uncertainty. **Uncertainty:** report source age, clock quality and client-driver variation, route timing, authentication cache and unmeasured handoff points separately. **Safe failure:** state that roaming is unverified for the route or device class when repeat tests disagree or route coverage is incomplete.

Authoritative supporting data: Owners: Wireless network operations and the owner of each operational route requirement. Precedence: the approved WLAN design, roaming policy, authentication method, test route and acceptance threshold define authorised intent; current controller state and controlled test-client evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate a route result after AP, authentication, firmware, partition or route change and record device model, software and test conditions. Conflicts: Expose route geometry, AP sequence, threshold, authentication policy and device-profile conflicts; do not choose silently.

Answer boundary: Must not track real clients, expose identities or credentials, promise uninterrupted sessions for all devices or alter radio, authentication or mobility settings.

Why this is useful: The team can reproduce the fault on a known route, distinguish RF from authentication delay and decide whether to tune, resurvey or escalate through authorised control.

IT-024

STAKEHOLDER QUESTION

Can the wireless design support the expected high-density teaching, event or public-use scenario in each relevant zone?

Sensors and telemetry: Proposed read-only evidence: AP radio capacity, aggregate association counts, airtime utilisation, retries, channel overlap and controlled synthetic load tests across approved high-density zones and the APs that serve them. **Coverage/timing:** Capture 5-minute controller aggregates, event-period summaries and bounded load tests only in an authorised window; summarise per-radio airtime, active-device ranges, retry rates and synthetic transaction performance. **Quality/provenance:** Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. **Limits:** planned attendance does not equal active devices; client mix, traffic demand, neighbouring cells and controller sampling can materially change capacity. Use aggregate counts and synthetic load only; do not identify attendees, inspect traffic or infer individual participation.

Analysis required: Type: estimate + scenario test. Decision interval: during event planning, after design change and after a high-density service issue. Minimum evidence: verified AP membership, a documented device and demand scenario, current radio health and either comparable event evidence or a permitted load test. Method: model conservative airtime, compare with measured retry and transaction behaviour, and test sensitivity to device count and traffic mix. **Uncertainty:** report source age, clock quality and scenario demand, client mix, airtime behaviour and untested event layouts separately. **Safe failure:** return a qualified scenario range and require a controlled test when the scenario lies outside observed evidence.

Authoritative supporting data: Owners: Wireless network operations, event or teaching service owners and facilities space-data owners. Precedence: the approved design capacity, zone geometry, expected technical demand and acceptance criteria define authorised intent; current controller state, historical aggregate event metrics and controlled test results show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat evidence as stale after AP, channel, antenna, furniture, partition or expected-device-profile changes. Conflicts: Expose zone capacity, AP membership, expected device count, traffic profile and event-period conflicts; do not choose silently.

Answer boundary: Must not guarantee participant experience, infer attendance from associations, conduct disruptive load generation without approval or change WLAN settings.

Why this is useful: The team can add capacity, alter the supported event design, narrow the commitment or commission a measured load test before service is promised.

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Wired, wireless and building-network coverage

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IT-025

STAKEHOLDER QUESTION

Which access switches or stacks lack verified PoE headroom for current devices or planned additions under startup, redundancy and failed-PSU conditions?

Sensors and telemetry: Proposed read-only evidence: per-port PoE draw, negotiated class, switch or stack power-supply state, PSU input state and connected-device inventory across authorised PoE switches serving access points, phones, cameras and building devices. Coverage/timing: Capture 30-60 second power samples with event capture for device start, PSU loss and stack-member change; summarise current, peak and worst-credible PoE demand by switch, stack and PSU state. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: startup surges may be missed, negotiated class may exceed or understate actual draw, and undocumented injectors or splitters break the model. Restrict device and switch detail by role; collect no media, call or sensor content from powered devices.

Analysis required: Type: calculation + recommendation. Decision interval: before adding devices, after PSU alarms and weekly for constrained stacks. Minimum evidence: verified port-device mapping, current PSU state, observed peak or manufacturer startup demand and the approved redundancy rule. Method: sum conservative demand, test normal and credible failed-PSU cases, and keep reserved margin separate from apparent spare budget. Uncertainty: report source age, clock quality and startup draw, PSU failure mode, cable loss and controller-reporting limits separately. Safe failure: state that PoE headroom is unverified when device demand, PSU mapping or startup evidence is missing.

Authoritative supporting data: Owners: Network operations, facilities where building devices are involved, and the relevant device owner. Precedence: approved switch design, PSU inventory, redundancy rule, PoE policy and device power requirements define authorised intent; current switch PoE telemetry, PSU state and verified device attachment show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Recalculate immediately after PSU, stack, firmware, port or device change and mark any unmapped powered device as an uncertainty. Conflicts: Expose device class, port mapping, PSU capacity, redundancy rule and measured-versus-rated draw conflicts; do not choose silently.

Answer boundary: Must not approve electrical capacity, change PoE state, cycle devices, alter switch configuration or promise that a device will start successfully.

Why this is useful: The team can select a viable switch or identify a PSU, cabling or design change before adding a device that would remove redundancy or trigger power denial.

IT-026

STAKEHOLDER QUESTION

Are managed, guest and building-service wireless networks available only in authorised zones with the expected authentication and upstream service?

Sensors and telemetry: Proposed read-only evidence: controller SSID and radio state, dedicated test-client association and synthetic service checks, and approved AP-to-zone mappings across zones where each wireless service is permitted or explicitly excluded. Coverage/timing: Capture controller state every 5 minutes and synthetic tests from fixed or controlled mobile test points every 5-15 minutes; summarise availability, authentication stage and upstream reachability by service and zone. Quality/provenance: Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: hidden SSIDs, captive portals, RF leakage across boundaries, stale zone mappings and test-client policy differences can affect interpretation. Use synthetic identities and test traffic; do not collect real client identities, credentials, payloads or browsing destinations.

Analysis required: Type: observation + comparison. Decision interval: daily for fixed tests, after policy or AP changes and during access incidents. Minimum evidence: a current service policy, verified zone, healthy test point and a permitted synthetic identity for the relevant network. Method: compare advertised service, authentication result and bounded upstream test with the policy matrix, treating RF presence and authorised usability separately. Uncertainty: report source age, clock quality and SSID visibility, authentication state, zone mapping and upstream-service dependence separately. Safe failure: return the zone-service pair as unknown when no current test point or valid identity exists.

Authoritative supporting data: Owners: Wireless operations, cybersecurity, identity owners and relevant building-service owners. Precedence: approved SSID policy, authentication method, zone authorisation and allowed upstream services define authorised intent; current controller configuration state and controlled test-client observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat zone or policy mappings as stale after AP, partition, identity-policy or room-use changes and retain the test identity and method version. Conflicts: Expose SSID name, zone boundary, authentication method, service reachability and policy-version conflicts; do not choose silently.

Answer boundary: Must not reveal credentials, bypass captive portals, test unauthorised networks, grant access, change SSIDs or assert that every real client can connect.

Why this is useful: The team can detect a missing service, policy leak or upstream dependency problem and route the action to WLAN, identity or cybersecurity owners with bounded evidence.

3

Wired, wireless and building-network coverage

IT-021 to IT-030 | sheet 4 of 5

IT-027

STAKEHOLDER QUESTION

Which approved plant, sensor, access-control or building-management endpoints have lost expected connectivity, and is the fault local, segment-wide or upstream?

Sensors and telemetry: Proposed read-only evidence: switch-port and controller state, NAC admission outcome, gateway heartbeat, bounded protocol-level health checks and approved device-service heartbeats across authorised building-service network segments and mapped endpoints. Coverage/timing: Capture link and heartbeat state every 1-5 minutes with immediate fault events and 15-minute segment summaries; summarise endpoint, access-switch, segment and gateway health without device payloads. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: sleep schedules, maintenance, local device failure, protocol gateways and owner-managed polling can create apparent loss; a reachable endpoint may still be functionally degraded. Do not collect building sensor values, camera feeds, access events, identities or control commands unless separately authorised for another purpose.

Analysis required: Type: diagnosis support. Decision interval: live incident triage and daily exception assessment. Minimum evidence: a verified endpoint mapping, expected connectivity window and at least two observations across endpoint, access and upstream layers. Method: compare peers on the same switch and segment, test the gateway from an approved point and classify local, shared or unobserved failure. Uncertainty: report source age, clock quality and endpoint sleep, controller blind spots, shared upstream faults and stale expected-state records separately. Safe failure: state "connectivity loss location unverified" when endpoint schedule or intermediate evidence is missing.

Authoritative supporting data: Owners: The building-service owner, network operations, cybersecurity and facilities. Precedence: the approved endpoint inventory, expected connectivity schedule, segment policy and maintenance state define authorised intent; current port, NAC, gateway and heartbeat observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply the endpoint's expected reporting interval and maintenance status; treat an unmapped or overdue heartbeat as unknown until its owner confirms context. Conflicts: Expose endpoint identity, expected schedule, segment, owner, maintenance and heartbeat-source conflicts; do not choose silently.

Answer boundary: Must not operate building systems, infer safety status from network reachability, inspect sensor or access data, restart gateways or change segmentation.

Why this is useful: The team can send the incident to the endpoint owner, access network or upstream service with fewer unnecessary site visits and without confusing connectivity with functional safety.

IT-028

STAKEHOLDER QUESTION

Do entrances, circulation routes, service routes and incident-command locations have verified connectivity where current policy requires it?

Sensors and telemetry: Proposed read-only evidence: fixed test nodes where justified, controlled mobile wired or wireless tests, AP telemetry and current outlet mappings across the exact operational spaces and routes named by service, safety or continuity owners. Coverage/timing: Capture fixed tests every 5-15 minutes and controlled route checks after material change or at the approved interval; summarise minimum required wired or wireless service by route segment and operational point. Quality/provenance: Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: doors, crowds, temporary barriers, weather exposure, room changes and test-device limits can alter coverage; fixed nodes cannot represent every point. Do not track people, devices or movements; use commissioned test equipment and aggregate service results only.

Analysis required: Type: lookup + observation. Decision interval: before an operational plan is relied on, after spatial or network change and during continuity preparation. Minimum evidence: a current authorised route or point, explicit connectivity requirement and a valid test within the relevant operating condition. Method: map only measured segments, distinguish continuous from spot evidence and report environmental or access conditions with each result. Uncertainty: report source age, clock quality and policy changes, route access, temporary obstructions and unmeasured operational positions separately. Safe failure: mark unmeasured or inaccessible segments unknown and escalate to the route owner for current confirmation.

Authoritative supporting data: Owners: Network operations, facilities, emergency or continuity owners and the relevant service owner. Precedence: current operational-route requirements, current authorised spatial records and connectivity acceptance criteria define authorised intent; commissioned fixed-node and controlled route-test observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Revalidate after route, partition, access, AP, outlet or service-policy change; the May 2021 floor plan cannot establish current route status or use. Conflicts: Expose route geometry, service requirement, test-point position, current access and threshold conflicts; do not choose silently.

Answer boundary: Must not certify an emergency route, promise communications availability, infer current access or hazards, track responders or change network configuration.

Why this is useful: The team can place a test node, schedule a route survey or provide an approved alternative connection before operational staff depend on an unverified location.

3

Wired, wireless and building-network coverage

IT-021 to IT-030 | sheet 5 of 5

IT-029

STAKEHOLDER QUESTION

For this failed wired connection, where is the earliest supported break in the outlet-to-upstream chain?

Sensors and telemetry: Proposed read-only evidence: controlled cable-certifier or TDR results, switch-port diagnostics, link and optics state, LLDP, patch identifiers and approved synthetic tests for the named outlet-to-upstream chain. Coverage/timing: capture live switch state and controlled tests within the authorised fault window; record evidence for each chain element. Quality/provenance: use an in-calibration instrument with the correct test limit and recorded endpoints; align instrument and work-record clocks, expose untested gaps, and retain source health, mapping drift and signed test lineage. Limits: TDR distance is approximate, live equipment may distort results, patch records may be stale and endpoint failure may imitate cable loss. Do not inspect endpoint content or use intrusive tests without authorisation and a safe method.

Analysis required: Type: diagnosis support. Decision interval: during an authorised fault investigation. Minimum evidence: a verified chain map plus current observations on both sides of a suspected break, or a controlled cable test. Method: test from the least intrusive layer outward, find the first transition from supported to failed or unknown, and preserve alternatives. Uncertainty: report clock quality, intermittency, access limits, undocumented patching and shared endpoint defects separately. Safe failure: stop at the first unverified element and request the appropriate physical or network test.

Authoritative supporting data: Owners: Network operations, authorised cabling specialists and the endpoint owner. Precedence: current outlet-to-port mapping, cable certification, switch design and approved test procedure define authorised intent; current port diagnostics, controlled cable tests and bounded synthetic connectivity observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use only evidence from the current fault window and reset the chain after repatching, cable replacement or port change. Conflicts: Expose outlet label, patch position, cable length, switch port, endpoint and test-method conflicts; do not choose silently.

Answer boundary: Must not direct unsafe work, test live circuits without authority, enable or reset ports, diagnose a user device or claim a cable defect without competent evidence.

Why this is useful: The team can send the right specialist and avoid replacing switches, cables or endpoints before the evidence has isolated the part of the chain that actually needs attention.

IT-030

STAKEHOLDER QUESTION

After a partition, furniture, room-use, route or equipment-layout change, which wired and wireless assumptions require fresh measurement?

Sensors and telemetry: Proposed read-only evidence: approved current alteration geometry, commissioned AP and outlet locations, prior survey cells and targeted post-change tests across the changed space and adjacent RF, cabling and operational-route influence area. Coverage/timing: Capture comparison at design approval and targeted measurements after the physical change is complete; summarise affected AP cells, outlet mappings, routes and previously supported coverage claims. Quality/provenance: Use a survey client with current calibration or reference verification and record antenna orientation, height and route. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: change drawings may differ from as-built conditions, materials may be unknown and RF or access effects can extend beyond the edited room. Use spatial and network asset data only; do not infer current room use from historical plans or monitor people.

Analysis required: Type: comparison + recommendation. Decision interval: at design, physical handover and before reopening or service acceptance. Minimum evidence: a defined change footprint, current asset map and the evidence date and method for each prior coverage claim. Method: expand the influence area using wall, route and RF adjacency, then list the minimum wired, RF, roaming and operational-point tests required. Uncertainty: report source age, clock quality and unrecorded spatial changes, furniture materials, new equipment and stale floor geometry separately. Safe failure: invalidate the affected claim and request current as-built evidence when geometry or completion status is uncertain.

Authoritative supporting data: Owners: Facilities space-data owners, project or works owners, network operations and affected service owners. Precedence: current approved as-built or handover records and the latest accepted network design or survey define authorised intent; commissioned asset locations and targeted post-change measurements show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. The May 2021 floor plan is historical reference only; treat any prior network claim within the influence area as stale until the change owner confirms completion and current geometry. Conflicts: Expose as-designed versus as-built geometry, AP or outlet position, access route and completion-date conflicts; do not choose silently.

Answer boundary: Must not certify as-built works, declare current room use, approve reopening, infer access or hazards, or retain old coverage claims as current fact.

Why this is useful: The team can scope a proportionate resurvey and prevent an apparently minor building change from silently invalidating outlet, Wi-Fi or operational-connectivity evidence.

4

Server-room power, cooling and environmental conditions

IT-031 to IT-040 | sheet 1 of 5

IT-031

STAKEHOLDER QUESTION

What is the current health, load, battery condition and alarm state of each infrastructure UPS, and which downstream assets depend on it?

Sensors and telemetry: Proposed read-only evidence: read-only UPS SNMP or Modbus state, input and output voltage, load, battery charge, battery temperature, self-test result, bypass state and mapped downstream PDU status across approved UPS units and protected communications or server loads. Coverage/timing: Capture 30-60 second samples with immediate alarm and state-change events plus daily summaries; summarise UPS health, load, protection mode and mapped service exposure. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: vendor metrics differ, branch mappings may be incomplete, battery state-of-charge is an estimate and monitor loss can look like a UPS failure. Restrict electrical and dependency detail by role; collect no user or application content.

Analysis required: Type: observation + calculation. Decision interval: live during alarms, daily for health and before electrical or infrastructure maintenance. Minimum evidence: a verified UPS identity, current telemetry, mapped protected loads and current maintenance or test status. Method: separate monitor failure from UPS alarm, compare load with approved operating limits and map confirmed versus assumed downstream exposure. Uncertainty: report source age, clock quality and battery-model accuracy, downstream mapping, self-test age and load transients separately. Safe failure: return protection state as unknown and escalate to facilities or the competent electrical owner when telemetry, mapping or safety context is incomplete.

Authoritative supporting data: Owners: Facilities, competent electrical professionals, infrastructure operations and UPS maintenance owners. Precedence: approved single-line diagrams, maintenance records, battery inventory, alarm definitions and protected-load mappings define authorised intent; current UPS, PDU and downstream infrastructure observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a UPS state older than two intervals as stale during an event and battery condition as stale after missed maintenance or self-test; historical spatial records do not establish current power routes. Conflicts: Expose UPS identity, branch mapping, battery age, alarm meaning, bypass state and downstream-load conflicts; do not choose silently.

Answer boundary: Must not operate bypasses or breakers, silence alarms, certify battery condition, instruct electrical intervention, promise runtime or authorise access to electrical equipment.

Why this is useful: The team can identify which services need continuity action, avoid unnecessary server work when the issue is upstream power and prepare a precise facilities escalation.

IT-032

STAKEHOLDER QUESTION

Are rack-inlet temperature and relative humidity within the approved envelope at the top, middle and bottom of each critical rack?

Sensors and telemetry: Proposed read-only evidence: calibrated temperature and humidity sensors mounted at commissioned rack-inlet positions, with local sensor health and battery or power state across critical network and server racks in authorised communications and server rooms. Coverage/timing: Capture 30-60 second samples with 5-minute averages, maxima, duration-above-limit and rate-of-change summaries; summarise conditions by rack height and inlet position rather than a single room average. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: poor placement, radiant heat, open rack doors, sensor drift, battery failure and missing top or bottom sensors can hide local exposure. Environmental evidence contains no user content; restrict rack location detail to authorised roles.

Analysis required: Type: observation + comparison. Decision interval: live during excursions, daily for exceptions and weekly for trend action. Minimum evidence: at least two healthy commissioned inlet sensors per critical rack, including the hottest credible height, plus current threshold and calibration status. Method: calculate duration and rate of excursion, compare top-to-bottom gradient and distinguish room-wide from rack-local change. Uncertainty: report source age, clock quality and sensor placement, calibration, rack airflow variation and missing vertical coverage separately. Safe failure: state that rack condition is unverified when placement, calibration or height coverage is inadequate.

Authoritative supporting data: Owners: Facilities, infrastructure operations and environmental-monitor owners. Precedence: the approved rack operating envelope, sensor-placement plan, calibration schedule and cooling-zone map define authorised intent; current commissioned rack-sensor observations and cooling state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat data as stale after sensor movement, rack rearrangement or missed calibration, and unknown after more than two missed samples during a thermal event. Conflicts: Expose rack identity, sensor position, threshold, calibration and cooling-zone conflicts; do not choose silently.

Answer boundary: Must not certify environmental compliance, change cooling, power down equipment, direct unsafe room entry or infer hidden equipment condition from air temperature alone.

Why this is useful: The team can prioritise the rack and height that needs inspection, check cooling or airflow before equipment alarms appear and avoid relying on a misleading room sensor.

4

Server-room power, cooling and environmental conditions

IT-031 to IT-040 | sheet 2 of 5

IT-033

STAKEHOLDER QUESTION

Where are persistent hot spots, exhaust recirculation or abnormal front-to-rear temperature differences developing?

Sensors and telemetry: Proposed read-only evidence: rack-inlet and outlet temperatures, optional commissioned differential-pressure sensors, cooling supply and return temperatures, and controlled thermal-survey observations across critical rack rows and cooling paths in authorised rooms. Coverage/timing: Capture 30-60 second fixed-sensor data with 5-minute aggregates and targeted thermal surveys after material layout or cooling change; summarise persistent spatial gradients, recirculation indicators and survey-confirmed hot cells. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: sparse sensors, temporary open doors, missing blanking panels, survey emissivity and air-mixing effects limit spatial interpretation.

Analysis required: Type: observation + comparison. Decision interval: live for severe excursions, weekly for emerging patterns and after physical changes. Minimum evidence: multiple healthy sensors spanning the relevant row or a current controlled survey plus known cooling operating state. Method: identify repeated gradients rather than single peaks, compare inlet and outlet behaviour, and limit interpolation to measured areas. Uncertainty: report source age, clock quality and sensor density, transient workloads, containment changes and local airflow disturbance separately. Safe failure: mark suspected hot zones as provisional and request a targeted survey when fixed-sensor density is insufficient.

Authoritative supporting data: Owners: Facilities cooling owners, infrastructure operations and authorised thermal-survey specialists. Precedence: the approved cooling design, rack layout, containment arrangement and sensor-placement plan define authorised intent; current rack, cooling and controlled survey observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate spatial conclusions after rack, containment, tile, fan or cooling changes and record survey method, date and operating condition. Conflicts: Expose rack layout, airflow direction, sensor position, cooling state and survey-to-fixed-sensor conflicts; do not choose silently.

Answer boundary: Must not diagnose concealed fire or electrical hazards, certify airflow design, alter cooling or containment, or direct staff into a potentially unsafe room.

Why this is useful: The team can target blanking, cable-management, containment or cooling checks and avoid treating a local recirculation problem as a whole-room capacity failure.

IT-034

STAKEHOLDER QUESTION

What cooling capacity and redundancy are currently evidenced for each critical rack zone, and which services are exposed by each approved single-failure scenario?

Sensors and telemetry: Proposed read-only evidence: read-only BMS cooling-unit state, supply and return temperatures, fan or compressor state, alarm status and rack-inlet temperatures across the cooling units, zones and racks that serve authorised communications and server rooms. Coverage/timing: Capture 30-60 second state with immediate alarms and 5-minute thermal summaries; summarise available cooling paths, current load proxy and rack exposure by zone. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: BMS points may be mapped incorrectly, cooling capacity is not directly measured by simple temperatures, shared plant and manual fallback can be hidden. Do not collect unrelated building-control data and restrict plant and room topology by role.

Analysis required: Type: calculation + estimate. Decision interval: during cooling alarms, before maintenance and quarterly for critical rooms. Minimum evidence: verified unit and zone mappings, current plant state, rack thermal evidence and an approved redundancy rule. Method: compare available paths with intended redundancy, simulate one named failure at a time and report exposed racks with uncertainty. Uncertainty: report source age, clock quality and rated versus delivered capacity, control-state visibility, shared cooling and failure-mode assumptions separately. Safe failure: return the failure consequence as unverified and escalate to the competent facilities owner when capacity or control behaviour is not evidenced.

Authoritative supporting data: Owners: Facilities cooling owners, competent HVAC professionals and infrastructure operations. Precedence: the approved cooling design, redundancy requirement, control sequence, maintenance state and rack-to-zone mapping define authorised intent; current BMS state, rack temperatures and authorised operator confirmation show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat plant status as stale after missed BMS updates or maintenance changes and rack-to-zone mapping as stale after layout or airflow changes. Conflicts: Expose unit state, capacity assumption, zone mapping, redundancy mode and maintenance conflicts; do not choose silently.

Answer boundary: Must not operate HVAC, override controls, certify cooling capacity or redundancy, promise safe runtime or authorise continued equipment operation.

Why this is useful: The team can escalate cooling impairment, protect exposed services, defer new load or plan controlled reduction using competent facilities evidence.

4

Server-room power, cooling and environmental conditions

IT-031 to IT-040 | sheet 3 of 5

IT-035

STAKEHOLDER QUESTION

Is there current evidence of water ingress or leak risk near racks, cabling, cooling or power distribution, and where is coverage absent?

Sensors and telemetry: Proposed read-only evidence: commissioned leak-detection cable or point sensors, condensate and drip-tray alarms, relevant BMS alarms and controlled visual confirmation records across known pipe, drain, cooling and water-entry routes near authorised communications and server infrastructure. Coverage/timing: Capture event-driven alarms with heartbeat every 1-5 minutes and scheduled controlled checks; summarise alarm location, coverage zone, sensor health and proximity to protected assets. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: dry sensors cannot prove absence outside their coverage, leaks can bypass sensing, cables can be displaced and contamination or condensation can cause false alarms. Do not capture room imagery by default; keep exact sensitive asset and water-route detail role restricted.

Analysis required: Type: observation. Decision interval: immediately on alarm, after water-related works and at the approved test interval. Minimum evidence: a healthy commissioned sensor or controlled confirmation, current asset proximity and the authorised response procedure. Method: distinguish alarm, sensor fault and uncovered area, identify the nearest protected assets without inferring water spread beyond evidence. Uncertainty: report source age, clock quality and sensor placement, inaccessible voids, contamination, battery state and uninstrumented routes separately. Safe failure: escalate as an unverified water risk and avoid a reassurance statement when sensor health or coverage is inadequate.

Authoritative supporting data: Owners: Facilities, competent building-services professionals, infrastructure operations and leak-system maintainers. Precedence: current pipe and drainage records, sensor coverage plan, maintenance records and response procedure define authorised intent; current leak-sensor heartbeat, alarm state, BMS evidence and authorised site confirmation show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a silent sensor without a current heartbeat as unknown and invalidate coverage after room, pipe, cooling or sensor-position changes; historical plans cannot establish current water routes. Conflicts: Expose sensor location, coverage zone, pipe route, alarm state and site-confirmation conflicts; do not choose silently.

Answer boundary: Must not declare an area safe, direct entry into a potentially hazardous space, isolate power, operate valves or certify absence of water.

Why this is useful: The team can protect service continuity, stop non-essential access through authorised channels and give facilities an exact sensor, zone and asset-impact starting point.

IT-036

STAKEHOLDER QUESTION

Did mains, UPS or PDU power-quality events coincide with infrastructure instability, and which alternative explanations remain credible?

Sensors and telemetry: Proposed read-only evidence: certified power-quality meters at approved upstream, UPS input or output and PDU points, together with time-aligned device resets, link events and host hardware alarms across mapped electrical feeds supporting affected infrastructure. Coverage/timing: Capture sub-cycle event capture for sags, swells or interruptions plus 30-60 second RMS and frequency summaries; summarise power events and infrastructure instability on a common time base. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: meter placement, clock offset, transfer behaviour, local PSU buffering and unmapped feeds can hide or misalign events.

Analysis required: Type: comparison + diagnosis support. Decision interval: during power or infrastructure incidents and monthly for repeated events. Minimum evidence: a verified common feed, calibrated event evidence and infrastructure events with clock uncertainty smaller than the comparison window. Method: align event windows, compare affected and unaffected equipment on the same and different feeds, and list power and non-power explanations. Uncertainty: report source age, clock quality and event timestamp alignment, waveform coverage, equipment logs and coincident non-power faults separately. Safe failure: report only temporal association and request competent electrical or hardware testing when causal evidence is insufficient.

Authoritative supporting data: Owners: Facilities electrical owners, competent electrical professionals and network or platform operations. Precedence: approved electrical topology, meter placement, event thresholds and equipment feed mappings define authorised intent; current certified-meter events, UPS or PDU state and infrastructure fault observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a correlation as invalid when clocks are not aligned within the event duration or feed mappings have changed since commissioning. Conflicts: Expose feed mapping, meter time, event definition, transfer state and device-reset conflicts; do not choose silently.

Answer boundary: Must not diagnose the electrical supply, certify power quality, isolate circuits, restart equipment or present coincidence as proof of cause.

Why this is useful: The team can determine whether to involve facilities, a hardware vendor or network and platform specialists first and preserve the right high-resolution evidence.

4

Server-room power, cooling and environmental conditions

IT-031 to IT-040 | sheet 4 of 5

IT-037

STAKEHOLDER QUESTION

Which racks, PDU branches or redundant feeds lack headroom or have problematic phase or A/B-load imbalance?

Sensors and telemetry: Proposed read-only evidence: metered PDU branch, phase and outlet current, voltage and power; UPS load; breaker-state telemetry where approved; and verified outlet-to-asset mappings across authorised server and communications racks and their mapped feeds. Coverage/timing: Capture 30-60 second samples with 5-minute peaks and daily sustained-load summaries; summarise normal, peak and credible single-feed demand by rack, branch, phase and A/B path. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: unmetered outlets, poor outlet mapping, startup peaks, breaker characteristics and shared upstream loads can make apparent headroom optimistic. Restrict electrical topology and asset mappings by role; do not collect application or user data.

Analysis required: Type: calculation + recommendation. Decision interval: daily for constrained racks and before adding, moving or powering equipment. Minimum evidence: verified asset-to-outlet mapping, current ratings, representative peak demand and the approved failed-feed scenario. Method: apply authorised derating, compare sustained and peak load, test loss of either A or B path and report the limiting component. Uncertainty: report source age, clock quality and meter mapping, phase attribution, startup peaks and A/B-dependency uncertainty separately. Safe failure: state that safe headroom is not established and escalate for competent electrical assessment when any critical mapping or rating is missing.

Authoritative supporting data: Owners: Facilities electrical owners, competent electrical professionals and infrastructure operations. Precedence: approved ratings, derating rules, A/B design, branch and outlet mappings and planned-load assumptions define authorised intent; current PDU, UPS and outlet observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Recalculate after feed, outlet, PDU, breaker or equipment change and treat missing peak data or unverified mappings as a headroom uncertainty. Conflicts: Expose rating, derating, branch mapping, phase assignment, A/B connection and planned-load conflicts; do not choose silently.

Answer boundary: Must not certify electrical safety, move loads, switch feeds, operate breakers, approve an installation or instruct work on energised equipment.

Why this is useful: The team can select a viable feed, defer a planned load or request a controlled rebalance before apparently spare rack space creates an electrical single point of failure.

IT-038

STAKEHOLDER QUESTION

Are server-room environmental sensors correctly located, healthy, time-aligned and within their approved calibration or validation interval?

Sensors and telemetry: Proposed read-only evidence: sensor heartbeat, battery or supply state, firmware and clock status, commissioned position tags, cross-sensor comparison and controlled reference checks across temperature, humidity, leak, differential-pressure and other approved room or rack sensors. Coverage/timing: Capture health every 1-5 minutes, clock status daily and controlled reference checks at the approved interval; summarise sensor quality state by type, position, calibration and gap rate. Quality/provenance: Use sensors at commissioned rack inlet, exhaust or leak-risk positions, not an arbitrary room average, and verify calibration or reference checks after moves. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: nearby sensors can legitimately differ, self-reported health can be wrong, tags can be moved and cross-comparison cannot replace a traceable reference check. No people, images or unrelated building data are required.

Analysis required: Type: observation + comparison. Decision interval: continuously for health, monthly for quality and before relying on a high-consequence environmental result. Minimum evidence: verified sensor identity and position, current health, clock status, gap rate and valid calibration or controlled cross-check evidence. Method: score placement, recency, drift, agreement and uptime separately and propagate the weakest relevant factor into downstream uncertainty. Uncertainty: report source age, clock quality and placement drift, calibration status, battery health, clock offset and unrecorded sensor moves separately. Safe failure: exclude the sensor from decisive analysis and return the condition as unknown when suitability is not established.

Authoritative supporting data: Owners: Environmental-monitor owners, facilities and infrastructure operations. Precedence: the commissioned sensor register, placement plan, calibration or validation procedure and maintenance history define authorised intent; current heartbeat, clock, battery, position and cross-check observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Mark a sensor invalid after movement, overdue reference check, unexplained drift or health gap beyond its allowed cadence. Conflicts: Expose sensor identity, physical position, calibration date, clock source, threshold and health conflicts; do not choose silently.

Answer boundary: Must not self-certify calibration, silently substitute a nearby sensor, alter thresholds or claim a room is normal when its evidence source is unhealthy.

Why this is useful: The team can avoid acting on a drifting or misplaced sensor, prioritise maintenance and know which environmental conclusions must be withdrawn until evidence is restored.

4

Server-room power, cooling and environmental conditions

IT-031 to IT-040 | sheet 5 of 5

IT-039

STAKEHOLDER QUESTION

What conservative UPS runtime range is defensible for each protected service at the present load and battery condition?

Sensors and telemetry: Proposed read-only evidence: current UPS load, battery charge and temperature, battery age and string state, latest self-test, measured transfer events and mapped protected loads across approved UPS systems and the critical services they protect. Coverage/timing: Capture 30-60 second load and state data with event capture and maintenance updates; summarise runtime inputs by UPS, load scenario and protected-service group. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: UPS runtime displays are estimates; battery ageing, temperature, nonlinear loads, future load and untested strings create wide uncertainty. Restrict electrical and service dependency detail to authorised roles.

Analysis required: Type: calculation + estimate. Decision interval: during power events, before maintenance and quarterly for critical continuity plans. Minimum evidence: current mapped load, battery age and condition, valid discharge basis and an approved service sequence. Method: calculate lower and central ranges for current and credible peak load, include transfer and shutdown time, and expose the assumptions that dominate uncertainty. Uncertainty: report source age, clock quality and battery ageing, temperature, load forecast, inverter efficiency and sequence duration separately. Safe failure: state that runtime cannot be supported and use the continuity plan's conservative fallback when battery or load evidence is inadequate.

Authoritative supporting data: Owners: Facilities electrical owners, UPS maintainers, infrastructure operations and service continuity owners. Precedence: approved battery inventory, manufacturer or validated discharge curves, maintenance and test records, load mapping and shutdown or failover plan define authorised intent; current UPS telemetry and latest controlled test evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat runtime evidence as stale after battery, load, firmware or topology change and when the latest required test is overdue. Conflicts: Expose battery identity, curve version, load mapping, test date, service priority and shutdown-sequence conflicts; do not choose silently.

Answer boundary: Must not guarantee runtime, operate the UPS, initiate shutdown or failover, certify battery health or replace the authorised continuity decision.

Why this is useful: The team can compare the available evidence with the time needed for controlled failover or shutdown and escalate before the remaining margin becomes unsafe.

IT-040

STAKEHOLDER QUESTION

How stable are power and environmental conditions across normal teaching, quiet periods, maintenance windows and exceptional operations?

Sensors and telemetry: Proposed read-only evidence: time-aligned rack temperature and humidity, cooling state, UPS and PDU load, power events and approved operational-calendar markers across critical communications and server spaces and their mapped infrastructure. Coverage/timing: Capture 30-60 second telemetry aggregated to 5-minute, hourly and operating-period summaries over 30-90 days; summarise condition distributions, excursions and variability by declared operating mode. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: weather, room access, temporary work, equipment changes and unequal period length can confound comparison; calendar labels may not reflect actual technical demand. Do not infer occupancy, attendance or staff behaviour from infrastructure or environmental patterns.

Analysis required: Type: comparison. Decision interval: monthly, seasonally and after an exceptional event. Minimum evidence: at least three comparable periods per mode where feasible, valid sensor coverage and current operating-mode labels. Method: compare robust distributions, duration above limits and variability while controlling for known configuration and external-condition changes. Uncertainty: report source age, clock quality and seasonal conditions, operating mode, missing high-resolution periods and changing rack load separately. Safe failure: report the comparison as inconclusive when modes, sensors or sample sizes are not comparable.

Authoritative supporting data: Owners: Facilities, infrastructure operations and service owners. Precedence: approved environmental limits, operating-mode definitions, maintenance windows and exceptional-event records define authorised intent; current time-series telemetry and authorised event markers show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Exclude intervals with changed sensors, rack layout or cooling configuration unless separately modelled and retain missing-data rates for every period. Conflicts: Expose operating-mode, event timing, threshold, sensor scope and configuration-change conflicts; do not choose silently.

Answer boundary: Must not infer people's presence or productivity, certify environmental compliance, attribute causation from correlation or control power and cooling.

Why this is useful: The team can identify conditions that only emerge during maintenance or peak technical demand and schedule a focused facilities or capacity intervention.

5

Alerts, incidents, dependencies and service restoration

IT-041 to IT-050 | sheet 1 of 5

IT-041

STAKEHOLDER QUESTION

Which active infrastructure alerts require immediate authorised action, which are likely symptoms, and which cannot be ranked because evidence is weak?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: read-only alert-manager events, monitor health, asset and service criticality, dependency tags, maintenance state and acknowledgement status across approved network, compute, storage, power, cooling and shared-service monitors.

Coverage/timing: Capture event-driven ingestion with second-level timestamps and 1-minute grouping windows; summarise deduplicated alert clusters with source, age, criticality and evidence completeness.

Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: suppression errors, duplicate collectors, stale dependencies, inconsistent severities and clock drift can distort priority. Use alert metadata only; do not expose log payloads, credentials, user content or sensitive topology beyond authorised roles.

Analysis required: Type: observation + recommendation. Decision interval: continuously during operations and at incident opening. Minimum evidence: a current alert, healthy source, mapped asset or service and at least one corroborating state for high-consequence prioritisation. Method: group by time, dependency and symptom, separate primary candidates from consequences, and keep unsupported critical alerts visible. Uncertainty: report source age, clock quality and monitor health, alert suppression, dependency context and severity-definition drift separately. Safe failure: return an unranked critical alert and escalate when mapping, source health or safety context is inadequate.

Authoritative supporting data: Owners: Incident management, domain operations, service owners and facilities for relevant alerts. Precedence: approved alert definitions, service criticality, maintenance state and incident-severity rules define authorised intent; current alert events, monitor health and dependency observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat an alert without current source health or outside its valid lifetime as unverified; preserve suppressed and cleared states with their times. Conflicts: Expose severity, asset identity, dependency, maintenance, duplicate-source and clock conflicts; do not choose silently.

Answer boundary: Must not acknowledge, suppress or close alerts, declare root cause, operate infrastructure, assign an incident severity outside policy or replace the duty manager.

Why this is useful: The team can set a defensible triage order and escalate uncertain high-consequence signals without suppressing them as mere noise.

IT-042

STAKEHOLDER QUESTION

What service, infrastructure and location blast radius is confirmed for the current incident, and which impacts remain only possible?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: current service health, dependency-graph state, network or hosting topology, power and cooling alarms and bounded synthetic checks across the incident assets, services and authorised building zones connected to them. Coverage/timing:

Capture 1-minute state during the incident with event updates and on-demand synthetic confirmation; summarise confirmed, probable and unobserved impact by service and zone. Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: incomplete dependencies, hidden supplier paths, stale room mappings and cascading effects can expand or shrink the apparent radius. Restrict sensitive topology and location detail to incident roles and collect no user content.

Analysis required: Type: lookup + calculation. Decision interval: at incident opening and whenever a key dependency or symptom changes. Minimum evidence: a verified initiating asset or service, current dependency graph and at least one current impact observation. Method: traverse only active or plausible dependency edges, distinguish confirmed failure from exposure, and show visibility boundaries and uncertainty. Uncertainty: report source age, clock quality and service mapping, user-journey coverage, incident evolution and location attribution separately. Safe failure: return the radius as partial and direct checks to the missing owner rather than treating every dependent as failed.

Authoritative supporting data: Owners: The incident commander or manager, service owners, infrastructure operations and facilities where relevant. Precedence: approved service dependency maps, current spatial records, criticality definitions and incident scope define authorised intent; current health, alert and synthetic-test observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. During an incident, treat dependency or spatial evidence older than the fault window as historical; the May 2021 floor plan cannot establish current room use or access. Conflicts: Expose service dependency, zone mapping, health state, ownership and incident-scope conflicts; do not choose silently.

Answer boundary: Must not disclose sensitive topology to unauthorised users, announce service impact publicly, infer safety status, authorise closures or control affected systems.

Why this is useful: The team can notify the right service owners, protect unaffected areas from unnecessary intervention and choose where to test next as the incident evolves.

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Alerts, incidents, dependencies and service restoration

IT-041 to IT-050 | sheet 2 of 5

IT-043

STAKEHOLDER QUESTION

Which shared dependency best explains several simultaneous failures, and what credible alternatives still need testing?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: time-aligned service failures, interface and device state, shared-service checks, power or cooling alarms, topology and hosting dependencies across the affected services and their authorised shared infrastructure. Coverage/timing: Capture event-driven state with 1-minute aggregation during the incident; summarise candidate shared dependencies and contradictory observations. Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: common timing can arise from monitoring loss, planned work, unrelated faults or a shared external provider; dependency maps may be incomplete. Use technical event and health metadata only; do not inspect user transactions or content.

Analysis required: Type: diagnosis support. Decision interval: during active incident diagnosis and after a material new observation. Minimum evidence: at least two affected services, a current dependency view and one discriminating observation for or against each leading candidate. Method: rank candidates by shared reach, timing and contradictions, then propose the least intrusive authorised test that separates them. Uncertainty: report source age, clock quality and dependency completeness, simultaneous independent faults, event timing and hidden supplier causes separately. Safe failure: state that no dependency is sufficiently supported and preserve the alternatives when evidence cannot discriminate.

Authoritative supporting data: Owners: Incident management, service owners and the relevant network, platform, storage, facilities or supplier owner. Precedence: approved dependency records and incident definitions define authorised intent; current alerts, health checks, topology state and authorised supplier evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Reject evidence with clock uncertainty larger than the observed sequence and mark stale dependency edges separately. Conflicts: Expose event time, service identity, shared-edge, maintenance and supplier-status conflicts; do not choose silently.

Answer boundary: Must not declare root cause, run intrusive tests, restart shared services, blame a supplier or conceal alternative explanations.

Why this is useful: The team can test the most discriminating dependency first and reduce repeated work across service teams without treating a plausible correlation as a proven cause.

IT-044

STAKEHOLDER QUESTION

When did the incident begin, what event sequence followed, and where do clock or collection gaps prevent a precise timeline?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: alert metadata, device and service state changes, monitoring health, approved change events, supplier incident times and infrastructure log-event headers across systems and services within the authorised incident scope. Coverage/timing: Capture event-driven capture retaining original timestamps, collector receipt times and clock-offset state; summarise a common timeline with source, confidence and gap markers. Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: device clocks, collector delays, duplicate events, retention loss and manual timestamps can disagree by seconds or minutes. Do not expose log payloads, user content, credentials or personal performance data; use only the event detail required for the incident.

Analysis required: Type: observation + timeline reconstruction. Decision interval: continuously during major incidents and at handover. Minimum evidence: original and receipt times for key events, current clock status and enough source metadata to identify duplicates. Method: convert to one time basis, represent uncertain events as intervals, order only when supported and mark collection gaps explicitly. Uncertainty: report source age, clock quality and clock offsets, collection delay, missing logs and inconsistent event granularity separately. Safe failure: leave ordering unresolved where uncertainty intervals overlap instead of inventing a precise sequence.

Authoritative supporting data: Owners: Incident management, domain operations, log-platform owners, change control and suppliers for their events. Precedence: the official incident record and approved time sources define authorised intent; original event headers, collector times, monitoring state and authorised operator entries show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Retain clock-offset and retrieval evidence; do not overwrite original times with a normalised value without preserving both. Conflicts: Expose event identity, timezone, clock source, duplicate, sequence and manually entered time conflicts; do not choose silently.

Answer boundary: Must not alter source logs, reconstruct user actions, attribute fault to a person, claim causal order from uncertain timestamps or certify forensic completeness.

Why this is useful: The team can coordinate parallel work, avoid false sequencing assumptions and hand over a timeline that shows exactly which gaps still need evidence.

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Alerts, incidents, dependencies and service restoration

IT-041 to IT-050 | sheet 3 of 5

IT-045

STAKEHOLDER QUESTION

Which alert rules create duplicate, stale or low-value noise without protecting visibility of genuine infrastructure risk?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: alert occurrence metadata, deduplication keys, source health, clear and acknowledgement times, incident linkage and maintenance context across approved infrastructure alert rules and monitors. Coverage/timing: Capture event-level capture with weekly and 30-90 day rule summaries; summarise frequency, duplicate ratio, persistence, incident linkage and missed-source-health context by rule. Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: an alert with few incident links may still be essential for rare high-impact events; acknowledgement patterns can reflect process rather than quality. Do not measure individual staff performance or expose alert payloads beyond the authorised operational need.

Analysis required: Type: comparison + recommendation. Decision interval: monthly and after an incident with material alert overload. Minimum evidence: a stable rule version, sufficient event volume or a documented rare-event purpose, and current service-owner input. Method: identify duplicate clusters, stale-after-clear behaviour and non-actionable repetition while preserving high-consequence exceptions. Uncertainty: report source age, clock quality and rare-event scarcity, suppression interaction, rule dependencies and changing service baselines separately. Safe failure: recommend no tuning when evidence cannot distinguish noise from necessary early warning.

Authoritative supporting data: Owners: Monitoring owners, domain operations, service owners and incident governance. Precedence: approved rule purpose, severity, dependency, maintenance behaviour and change procedure define authorised intent; current alert metadata and incident linkage show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Compare only periods with consistent rule definitions and monitor scope; retain rule version and effective dates. Conflicts: Expose rule version, severity, duplicate key, incident linkage and maintenance-classification conflicts; do not choose silently.

Answer boundary: Must not disable, suppress or alter rules, rank staff performance, hide critical alerts or approve a monitoring change.

Why this is useful: The team can take a focused, evidence-linked tuning proposal through change control while keeping rare but important failure signals protected.

IT-046

STAKEHOLDER QUESTION

After restoration work, what evidence shows the affected service and critical dependencies are stable rather than briefly reachable?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: synthetic service transactions, dependency health, error and latency rates, failover state, alert recurrence and monitor health across the restored service path and the dependencies named in the incident recovery criteria. Coverage/timing: Capture 1-minute observations through an agreed stabilisation period with event capture; summarise recovery criteria, recurrence count and remaining degraded state. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a short observation window, low demand, warm caches, partial client paths and hidden supplier components can create false confidence. Use synthetic tests and aggregate service state; do not inspect real user content or sessions.

Analysis required: Type: observation + comparison. Decision interval: continuously after restoration and at technical handback. Minimum evidence: successful checks across the critical path, healthy monitors, no unresolved high-severity dependency alarm and the full agreed observation window. Method: evaluate each criterion separately, compare with pre-incident baseline and keep residual risk or missing paths visible. Uncertainty: report source age, clock quality and short observation windows, partial journey tests, hidden dependencies and intermittent recurrence separately. Safe failure: return "restoration not yet evidenced" when the observation window or dependency checks are incomplete.

Authoritative supporting data: Owners: The incident manager, service owner and responsible infrastructure teams. Precedence: approved recovery criteria, dependency list, observation period and accepted degraded modes define authorised intent; current synthetic, monitoring and dependency observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the criteria and window effective for this incident; invalidate evidence after another material change or renewed alert. Conflicts: Expose acceptance threshold, test path, dependency state, observation-window and degraded-mode conflicts; do not choose silently.

Answer boundary: Must not close the incident, certify full service recovery, restart or fail over systems, promise user impact has ended or override the incident manager.

Why this is useful: The team can avoid premature closure, extend observation and hand back with passed, failed and still-unverified recovery conditions.

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Alerts, incidents, dependencies and service restoration

IT-041 to IT-050 | sheet 4 of 5

IT-047

STAKEHOLDER QUESTION

Did failover operate as designed, and are active and standby components within the approved degraded-state limits?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: cluster, routing, first-hop redundancy, load-balancer, storage-replication and orchestrator state plus bounded synthetic service tests across approved resilient infrastructure and the services that depend on it. Coverage/timing: Capture event-driven role changes with 30-60 second state and 1-minute service checks; summarise active role, standby readiness, replication state, capacity and service outcome. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: split-brain views, stale controllers, partial failover, hidden dependencies and low post-failover demand can mask risk. Restrict detailed resilience topology by role and collect no application payloads.

Analysis required: Type: observation + comparison. Decision interval: during failover, immediately after it and through the agreed stabilisation period. Minimum evidence: current role state from the authoritative control plane, service checks and evidence that the surviving path has sufficient capacity. Method: compare observed transition order and state with design, test service outcomes and identify lost redundancy or backlog. Uncertainty: report source age, clock quality and ambiguous active state, stale orchestration data, partial failover and degraded-capacity limits separately. Safe failure: report failover as partial or unverified and escalate rather than triggering additional transitions.

Authoritative supporting data: Owners: Network, platform, storage and service owners plus incident management. Precedence: approved failover design, role hierarchy, capacity limits, recovery sequence and accepted degraded mode define authorised intent; current control-plane, replication and synthetic-test observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat role or replication state older than two intervals as stale and reset the assessment after any manual intervention. Conflicts: Expose active-role, quorum, replication, service path, capacity and degraded-mode conflicts; do not choose silently.

Answer boundary: Must not initiate or reverse failover, force quorum, change routes, certify resilience or close the incident.

Why this is useful: The team can see whether service has truly transferred, which resilience layer is now depleted and what must be restored before the next failure can be tolerated.

IT-048

STAKEHOLDER QUESTION

Is a carrier, cloud, managed-platform or supplier incident contributing to the outage, and what local evidence supports escalation?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: local demarcation counters, bounded synthetic tests, DNS or route observations, supplier status feeds and authorised incident references across approved external service boundaries and local test points. Coverage/timing: Capture local state every 1-5 minutes, probes on demand and supplier updates at their published cadence; summarise local versus external evidence with first-observed and supplier-confirmed times. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: supplier status can lag, Internet paths can vary, multiple providers can be involved and a clean local hand-off does not prove remote failure. Do not probe supplier internals, collect user traffic or disclose contract-sensitive information outside authorised roles.

Analysis required: Type: lookup + diagnosis support. Decision interval: at incident triage and whenever supplier evidence changes. Minimum evidence: a verified boundary, healthy local test point, at least one local observation and a current supplier status or documented absence of one. Method: compare independent local paths where possible, identify the first unobserved external segment and package evidence without claiming remote root cause. Uncertainty: report source age, clock quality and supplier-status delay, boundary-test coverage, local coincidence and contract demarcation separately. Safe failure: state that supplier contribution is unconfirmed and continue authorised local checks when evidence does not isolate the boundary.

Authoritative supporting data: Owners: The service and contract owners, network or platform operations and the supplier. Precedence: the current service demarcation, support entitlement and escalation procedure define authorised intent; local tests, supplier status, incident IDs and authorised supplier responses show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply contract effective dates and distinguish supplier publication time from event time; treat generic public status as lower precedence than a case-specific response. Conflicts: Expose demarcation, incident ID, affected region, service scope, timestamp and supplier-response conflicts; do not choose silently.

Answer boundary: Must not blame a supplier, speak for them, promise restoration times or credits, perform intrusive external tests or authorise contractual action.

Why this is useful: The team can open or update the supplier case with precise times, boundary tests and affected services while avoiding repeated local checks that cannot see beyond the hand-off.

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Alerts, incidents, dependencies and service restoration

IT-041 to IT-050 | sheet 5 of 5

IT-049

STAKEHOLDER QUESTION

What evidence is still missing for a safe operational handover between shifts, technical teams or suppliers?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: incident-record fields, current alert and service state, timeline gaps, open actions, evidence links, owner roles and monitoring coverage across the authorised incident workspace and affected service scope. Coverage/timing: Capture continuous field updates with a completeness snapshot at each proposed handover; summarise required versus present handover elements without measuring individual performance. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: information may exist in restricted systems, verbal decisions may be unrecorded and rapidly changing state can make a snapshot outdated. Do not expose private contact details, credentials, log payloads or content outside the receiving role's permission.

Analysis required: Type: lookup + completeness check. Decision interval: before each shift, team or supplier handover. Minimum evidence: current incident status, affected services, actions completed and pending, residual risk, evidence links and named receiving role. Method: check mandatory fields, flag stale statements, distinguish fact from hypothesis and identify permissions that block evidence transfer. Uncertainty: report source age, clock quality and missing hypotheses, stale ownership, unrecorded actions and unresolved safety or access constraints separately. Safe failure: state that handover is incomplete and list the minimum missing items rather than filling gaps by inference.

Authoritative supporting data: Owners: Incident management, current and receiving duty roles, service owners and suppliers for their actions. Precedence: the approved handover template, incident severity procedure and role-based information requirements define authorised intent; current incident record, monitoring state and authorised operator entries show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Stamp the handover snapshot with observed and retrieved times and invalidate it after material incident-state change. Conflicts: Expose owner, action status, timeline, risk, evidence-link and receiving-role conflicts; do not choose silently.

Answer boundary: Must not accept the handover, assign responsibility, disclose restricted evidence, invent missing actions or replace the incident manager's judgement.

Why this is useful: The team can transfer control with fewer repeated diagnostics and make the unresolved risks, hypotheses and next authorised actions explicit to the receiving role.

IT-050

STAKEHOLDER QUESTION

Which recurring evidence-supported conditions across closed incidents justify a controlled preventive test, monitoring change or maintenance proposal?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: normalised incident metadata, affected services, infrastructure alerts, dependency state, approved change context, power or cooling events and restoration actions across closed incidents within the authorised infrastructure scope. Coverage/timing: Capture event-level evidence retained according to policy and analysed in monthly or quarterly aggregates; summarise recurring technical conditions and evidence coverage across comparable incidents. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: incident taxonomies change, small samples mislead, common symptoms need not share a cause and closed records may omit negative evidence. Exclude personal performance measures, user content and unnecessary raw logs.

Analysis required: Type: comparison + recommendation. Decision interval: monthly for high-frequency issues and quarterly for broader patterns. Minimum evidence: at least three comparable incidents or a documented high-consequence exception, with sufficient evidence to test the proposed common condition. Method: cluster by symptom and dependency, compare the condition against non-incident periods where possible, and separate accepted cause from recurring association. Uncertainty: report source age, clock quality and small incident samples, classification drift, changed architecture and survivor bias separately. Safe failure: return the pattern as provisional and recommend a bounded data collection or test when sample size or evidence quality is weak.

Authoritative supporting data: Owners: Incident governance, service owners and relevant infrastructure domains. Precedence: approved incident classification, problem records and accepted causal findings define authorised intent; time-stamped incident, monitoring and change evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Compare only records with compatible definitions and retain taxonomy version, evidence completeness and closure status. Conflicts: Expose incident class, affected scope, causal finding, evidence quality and action-outcome conflicts; do not choose silently.

Answer boundary: Must not blame individuals or suppliers, rewrite accepted incident findings, declare root cause from association or automatically create or approve preventive changes.

Why this is useful: The team can justify a specific monitoring, resilience, maintenance or configuration proposal with evidence from repeated incidents rather than relying on memory or anecdote.

6

Changes, maintenance windows and configuration evidence

IT-051 to IT-060 | sheet 1 of 5

IT-051

STAKEHOLDER QUESTION

What approved infrastructure maintenance is scheduled, which services and building functions may be affected, and where do technical or supplier windows conflict?

Sensors and telemetry: No new physical sensor is justified. **Proposed read-only evidence:** read-only change-calendar, maintenance-window, dependency, supplier-notice and current service-health integrations across approved network, platform, storage, power, cooling and supplier work relevant to the building. **Coverage/timing:** Capture calendar refresh on record change and a current health snapshot before each window; summarise time intervals, affected dependencies, owners and expected service states. **Quality/provenance:** Validate schema, effective dates, owner roles, record version and integration scope. **Record clock alignment, source health, gaps or drift, and source-to-summary lineage.** **Limits:** late approvals, informal work, stale dependencies, timezone differences and supplier updates can make the combined view incomplete. Do not expose confidential contract detail or personal rota information beyond authorised roles.

Analysis required: Type: lookup + comparison. **Decision interval:** during planning, at approval gates and before work starts. **Minimum evidence:** approved windows with start, end, owner, scope, expected impact and mapped dependencies. **Method:** detect direct and dependency-mediated overlaps, identify simultaneous loss of redundancy and show missing owner or supplier coverage. **Uncertainty:** report source age, clock quality and window time zones, dependency overlap, supplier constraints and late scope changes separately. **Safe failure:** return the combined schedule as incomplete and defer to change control when approval, timing or dependency evidence is missing.

Authoritative supporting data: **Owners:** Change control, maintenance owners, service owners, facilities and contract managers. **Precedence:** the approved change and maintenance records, service calendar, freeze periods and supplier notices define authorised intent; current health and dependency observations only for validating present context show current observation; an authorised owner or competent site check resolves exceptions. **Time/provenance:** store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use each record's effective time, status and timezone; provisional or expired windows must not be treated as approved. **Conflicts:** Expose window status, timezone, service impact, dependency, owner and supplier-notice conflicts; do not choose silently.

Answer boundary: Must not approve, reject, reschedule or cancel work, authorise access, promise no impact or override change freezes and supplier controls.

Why this is useful: The team can avoid overlapping maintenance on shared dependencies, coordinate test and staffing cover and present a complete impact picture to the authorised approver.

IT-052

STAKEHOLDER QUESTION

Did a recent approved, emergency or supplier change precede this fault on a relevant dependency path and therefore warrant focused checking?

Sensors and telemetry: No new physical sensor is justified. **Proposed read-only evidence:** approved change events, configuration-state hashes, topology or placement changes, monitoring transitions and incident symptoms across the affected service and its authorised dependency path. **Coverage/timing:** Capture event-driven records spanning a defined pre-incident and incident window; summarise candidate changes with timing, dependency distance and evidence quality. **Quality/provenance:** Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. **Record clock alignment, source health, gaps or drift, and source-to-summary lineage.** **Limits:** concurrent work, undocumented changes, delayed symptoms, monitor restarts and clock error can create false or missed associations. Use change metadata and sanitised state only; do not expose secrets or attribute action to individuals unnecessarily.

Analysis required: Type: comparison + diagnosis support. **Decision interval:** at incident triage and when new timeline evidence appears. **Minimum evidence:** a current incident start interval, candidate change actual time and a supported dependency path or state difference. **Method:** rank changes by temporal overlap, path relevance and matching symptom while preserving unrelated and undocumented-change alternatives. **Uncertainty:** report source age, clock quality and clock alignment, overlapping changes, latent faults and incomplete dependency mapping separately. **Safe failure:** state that no change is sufficiently linked and continue normal diagnosis when timing or dependency evidence is weak.

Authoritative supporting data: **Owners:** Change control, implementing teams, service owners and suppliers for their changes. **Precedence:** approved or emergency change records and the current dependency model define authorised intent; time-aligned telemetry, state hashes and incident observations show current observation; an authorised owner or competent site check resolves exceptions. **Time/provenance:** store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. **Retain planned, actual and observed times separately;** do not treat an approved record as proof the implementation occurred exactly as planned. **Conflicts:** Expose change scope, actual time, dependency, state difference and incident-start conflicts; do not choose silently.

Answer boundary: Must not declare a change as root cause, blame an implementer, roll back work, expose sensitive configuration or bypass emergency-change procedures.

Why this is useful: The team can check the most relevant change evidence first and avoid either ignoring a plausible trigger or reversing an unrelated change based only on timing.

6

Changes, maintenance windows and configuration evidence

IT-051 to IT-060 | sheet 2 of 5

IT-053

STAKEHOLDER QUESTION

Where does observed network, server, virtualisation or storage state differ from the approved baseline, and which differences are operationally material?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: approved read-only controller or platform state, canonical configuration properties, version identifiers and cryptographic hashes that exclude secrets across authorised infrastructure assets and configuration domains. Coverage/timing: Capture daily snapshots plus event-driven collection after approved configuration changes; summarise property-level intended, observed and unknown state with evidence age. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: vendor defaults, generated values, ordering, ephemeral state and unsupported collectors can create false differences. Do not collect passwords, keys, tokens, full sensitive configurations or user content; reveal only the minimum authorised difference.

Analysis required: Type: comparison. Decision interval: daily, after changes and during incident diagnosis. Minimum evidence: a verified asset, current approved baseline, current sanitised observation and a domain rule for material properties. Method: normalise vendor syntax, ignore approved ephemeral fields, classify expected, excepted, unexplained and unobservable differences, and link each to evidence. Uncertainty: report source age, clock quality and generated-configuration noise, baseline age, device-role changes and partial collection separately. Safe failure: return the property as unverified and request owner confirmation when collection or baseline compatibility is inadequate.

Authoritative supporting data: Owners: Configuration owners, network or platform operations and change control. Precedence: the approved baseline, template, policy or signed intended-state record define authorised intent; current sanitised controller, device or platform state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat a baseline as stale after a superseding approval and observed state as stale after missed collection or device replacement; retain schema and collector version. Conflicts: Expose asset identity, baseline version, property meaning, exception and collector-normalisation conflicts; do not choose silently.

Answer boundary: Must not disclose secrets or complete sensitive configurations, modify state, perform remediation, approve exceptions or claim that every difference is harmful.

Why this is useful: The team can focus on material drift, confirm benign generated differences and create a bounded change or incident task without manually comparing entire configurations.

IT-054

STAKEHOLDER QUESTION

Which firmware, operating systems, hypervisors or platform components are outside the approved support or patch position, and what service exposure follows?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: verified asset and component versions, vendor or internal lifecycle metadata, approved patch-catalogue state and service dependency mappings across authorised network, compute, storage and management infrastructure. Coverage/timing: Capture version collection daily or after change and lifecycle or patch feeds at their approved publication cadence; summarise component support, approved patch position and dependent-service exposure. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: embedded components, backported fixes, custom support, appliance bundles and incomplete version discovery can make simple version comparison wrong. Do not exploit systems, collect payloads or expose sensitive version and topology detail outside authorised cybersecurity and operations roles.

Analysis required: Type: lookup + estimate. Decision interval: weekly, after major advisories and before maintenance planning. Minimum evidence: verified product identity and version, current approved policy, applicable lifecycle or patch evidence and mapped service criticality. Method: match exact product context, account for accepted backports or exceptions, and rank by actual service and network position rather than version age alone. Uncertainty: report source age, clock quality and product identification, vendor applicability, backported fixes and support-data age separately. Safe failure: label support or patch status unknown and request product-owner validation when identity or vendor evidence is ambiguous.

Authoritative supporting data: Owners: Cybersecurity, product or platform owners, vendor-management owners and change control. Precedence: the approved patch policy, risk exceptions, vendor lifecycle and internally accepted support position define authorised intent; current verified version inventory and authorised vendor or internal advisories show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply advisory publication, withdrawal and exception effective dates; mark unknown bundle contents or unsupported discovery as uncertain. Conflicts: Expose version, product edition, backport status, exception, support entitlement and dependency conflicts; do not choose silently.

Answer boundary: Must not patch, reboot, exploit-test, certify security, disclose detailed exposure broadly or override cybersecurity and change decisions.

Why this is useful: The team can build a realistic maintenance queue, protect high-exposure services first and avoid false alarms caused by misidentified product or backported fixes.

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Changes, maintenance windows and configuration evidence

IT-051 to IT-060 | sheet 3 of 5

IT-055

STAKEHOLDER QUESTION

Do critical devices and platforms have current protected configuration backups whose integrity and restoration procedure have been proven?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: backup-job status, last-success time, protected repository state, object hash, encryption and key-escrow status indicators, and controlled restore-test outcomes across approved network, server, storage and management configurations. Coverage/timing: Capture job and repository state daily with event alerts and restore-test records at the approved interval; summarise coverage, freshness, integrity and proven-restoration state by asset or platform. Quality/provenance: Validate job identity, protected scope, repository, retention, checksums and restore-test evidence. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a successful job may capture the wrong scope, hashes do not prove semantic usability, and key or repository dependencies can fail independently. Do not retrieve or display backup content, credentials, private keys or sensitive configuration except through the authorised recovery procedure.

Analysis required: Type: lookup + observation. Decision interval: daily for failures, monthly for coverage and at continuity preparation. Minimum evidence: verified asset-to-backup mapping, current successful object, protected retention, integrity evidence and a valid controlled restoration result for critical classes. Method: separate job success, object integrity, protection and tested usability, and show dependencies needed to retrieve and apply the configuration. Uncertainty: report source age, clock quality and backup scope, protected-secret custody, restore dependencies and test-environment differences separately. Safe failure: return "recoverability unproven" when a valid restoration test or key dependency is missing.

Authoritative supporting data: Owners: Configuration owners, backup owners, cybersecurity and service continuity owners. Precedence: the approved backup scope, retention, protection, key-custody and restoration procedure define authorised intent; current job metadata, repository health, integrity checks and authorised restore-test evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply asset-specific freshness and test intervals; invalidate status after platform replacement, major configuration change or lost repository or key dependency. Conflicts: Expose asset scope, backup object, retention, integrity, key custody and restore-test conflicts; do not choose silently.

Answer boundary: Must not expose configuration content or keys, restore configurations, alter retention, certify recoverability or replace authorised recovery testing.

Why this is useful: The team can correct missing coverage or failed dependencies before an incident and distinguish a recent backup from one that is actually usable in the controlled recovery process.

IT-056

STAKEHOLDER QUESTION

Which observed configuration differences lack a matching approved change, active exception or documented emergency action?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: sanitised current configuration state, approved baseline hashes, change and exception metadata, and asset identity across authorised infrastructure assets and material configuration properties. Coverage/timing: Capture daily snapshots plus collection after approved or emergency changes; summarise unmatched property differences by asset, effective time and confidence. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: late records, failed collection, generated state, emergency work and asset replacement can create apparent unauthorised drift. Do not collect secrets, command history or personal attribution beyond what authorised investigation requires.

Analysis required: Type: comparison. Decision interval: daily and during incident or control validation. Minimum evidence: a current approved baseline, current observation and complete authorised records for the comparison window. Method: match each material difference to an approved change or exception, distinguish timing gaps from unexplained drift and preserve evidence links. Uncertainty: report source age, clock quality and emergency actions, collector gaps, automation-generated changes and undocumented exceptions separately. Safe failure: classify the difference as unexplained, not malicious, and escalate to the configuration owner when records or timing are incomplete.

Authoritative supporting data: Owners: Configuration owners, change control, cybersecurity and implementing teams. Precedence: the approved baseline, change records, active exceptions and emergency-action records define authorised intent; current sanitised device or platform state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use actual implementation and exception effective times, and mark state collected before record completion as time-uncertain. Conflicts: Expose asset identity, property, baseline, change scope, exception and observed-time conflicts; do not choose silently.

Answer boundary: Must not declare compromise, identify a person as responsible, expose secrets, revert state, create retrospective approval or close an investigation.

Why this is useful: The team can validate legitimate emergency work quickly and route genuinely unexplained drift through the correct configuration or cybersecurity process.

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Changes, maintenance windows and configuration evidence

IT-051 to IT-060 | sheet 4 of 5

IT-057

STAKEHOLDER QUESTION

Before this change is authorised, which service, dependency, capacity, resilience, access, observability or rollback risks still lack evidence?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: current service health, dependency maps, capacity and redundancy state, backup or rollback readiness, monitor coverage and authorised access prerequisites across the assets and services named in the proposed change scope. Coverage/timing: Capture a current pre-change snapshot within the approval-defined validity window; summarise precondition status and evidence gaps by change step. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: hidden dependencies, rapidly changing load, stale backups, supplier constraints and untested rollback can make a green snapshot misleading. Use operational metadata only and expose sensitive access or topology details solely to authorised change roles.

Analysis required: Type: recommendation. Decision interval: at technical design, approval and immediately before execution. Minimum evidence: approved scope and method plus current evidence for critical dependencies, capacity, redundancy, access, monitoring and rollback. Method: evaluate hard stop conditions separately from manageable risks, test loss of a remaining redundant component and list owner decisions required. Uncertainty: report source age, clock quality and unverified dependencies, capacity margins, rollback timing and monitoring blind spots separately. Safe failure: return the change readiness as incomplete and defer to authorised control when any hard precondition is unsupported.

Authoritative supporting data: Owners: The change owner, service owners, infrastructure domains, cybersecurity, facilities and change control. Precedence: the proposed method, acceptance and rollback plan, risk assessment, approved access and change policy define authorised intent; current operational observations and owner confirmations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Each precondition must carry an expiry time appropriate to its volatility; invalidate the snapshot after a material state change. Conflicts: Expose scope, dependency, capacity limit, redundancy, access, backup, supplier and rollback-readiness conflicts; do not choose silently.

Answer boundary: Must not approve the change, grant access, execute commands, waive risk, certify rollback or replace competent electrical, cybersecurity or service-owner judgement.

Why this is useful: The change owner can hold, narrow or reschedule the work, commission missing checks and present explicit residual risk to the authorised approver.

IT-058

STAKEHOLDER QUESTION

After maintenance, have intended configuration, service checks, redundancy, monitoring and alerting returned to the approved handback state?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: before-and-after sanitised state hashes, controller or platform health, synthetic service tests, redundancy state, monitor heartbeat and alert-path tests across the systems and services within the approved maintenance scope. Coverage/timing: Capture baseline before work, event capture during it and repeated tests through the agreed stabilisation period; summarise handback criterion results with evidence times and remaining degraded states. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: tests can miss client paths, monitoring may recover later than service, and temporary bypass or maintenance mode can hide faults. Do not expose secrets, use real user transactions or alter configuration and alerts during validation.

Analysis required: Type: observation + comparison. Decision interval: before technical handback and after the full stabilisation period. Minimum evidence: current intended-state comparison, successful critical service checks, restored monitoring and evidence for required redundancy. Method: evaluate each criterion independently, compare with the pre-work baseline and keep waived or untested items explicit. Uncertainty: report source age, clock quality and test coverage, delayed alerts, redundancy state and post-maintenance observation duration separately. Safe failure: return "handback criteria incomplete" and list the missing test or owner decision instead of assuming success.

Authoritative supporting data: Owners: The maintenance owner, service owners, monitoring owners and change control. Precedence: the approved implementation plan, expected state, acceptance criteria and stabilisation period define authorised intent; current sanitised state, synthetic tests, redundancy and monitor observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use criteria effective for this maintenance and invalidate the result after any subsequent change or renewed alert. Conflicts: Expose expected state, test scope, redundancy, monitor coverage, alert path and accepted-degradation conflicts; do not choose silently.

Answer boundary: Must not accept handback, close the change, remove maintenance mode, restart services, change alerts or promise that all users are unaffected.

Why this is useful: The team can catch forgotten monitoring, lost redundancy or partial service recovery before the maintenance team leaves and the next shift assumes normal operation.

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Changes, maintenance windows and configuration evidence

IT-051 to IT-060 | sheet 5 of 5

IT-059

STAKEHOLDER QUESTION

Which temporary exceptions, bypasses or workarounds are near expiry, ownerless, or no longer supported by working compensating controls?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: exception-register metadata, observed sanitised state, expiry dates, owner-role status, compensating-control health and linked incident or change records across approved temporary deviations affecting infrastructure. Coverage/timing: Capture daily state and expiry checks with event updates when ownership, control health or configuration changes; summarise active, expired, orphaned and state-mismatched exceptions. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: the register may lag emergency work, compensating controls may be outside monitoring and an expired date does not prove the configuration remains active. Do not expose sensitive workaround details broadly or collect personal performance information.

Analysis required: Type: lookup + comparison. Decision interval: daily for imminent expiry and monthly for the full set. Minimum evidence: a current exception record, verified affected state, accountable owner role and evidence for each required compensating control. Method: classify expiring, expired-active, expired-inactive, orphaned and control-unverified cases and rank by service exposure. Uncertainty: report source age, clock quality and exception ownership, expiry metadata, compensating-control health and hidden dependencies separately. Safe failure: escalate the exception as unresolved without extending, revoking or changing it when ownership or state cannot be verified.

Authoritative supporting data: Owners: Exception owners, cybersecurity, change control, service owners and relevant infrastructure teams. Precedence: the approved exception record, expiry, scope, risk acceptance and compensating-control requirement define authorised intent; current sanitised configuration and control-health observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply actual effective and expiry times and mark owner records stale when roles or employment status change. Conflicts: Expose scope, observed state, expiry, owner, risk acceptance and compensating-control conflicts; do not choose silently.

Answer boundary: Must not extend, revoke or approve exceptions, expose sensitive workaround configuration, perform remediation or infer acceptance from silence.

Why this is useful: The team can remove forgotten workarounds through authorised control, renew only those still justified and prevent a temporary weakened state from becoming permanent by default.

IT-060

STAKEHOLDER QUESTION

Is the evidence package for this completed infrastructure change sufficient for operational handback and later investigation?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: approved change fields, implementation timestamps, sanitised before-and-after state, test results, configuration-backup status, monitoring state and open defect references across the authorised change record and its affected assets and services. Coverage/timing: Capture evidence captured before, during and after the window with a final snapshot at proposed handback; summarise required evidence items, freshness and traceable source links. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: files can be inaccessible, tests can lack scope, manual notes can omit times and later work can overwrite the immediate post-change state. Do not copy credentials, keys, full sensitive configurations, user content or unnecessary personal details into the package.

Analysis required: Type: lookup + completeness check. Decision interval: at proposed technical handback and before record closure. Minimum evidence: approved scope, actual implementation record, state comparison, service and monitoring tests, rollback or backup evidence and known residual defects. Method: check that each required item is present, current, permission-accessible and linked to the correct asset and window. Uncertainty: report source age, clock quality and missing timestamps, incomplete command evidence, absent test results and unresolved exceptions separately. Safe failure: state that operational handback evidence is incomplete and list missing items without fabricating or substituting proof.

Authoritative supporting data: Owners: The change owner, implementing teams, service owners and change control. Precedence: the approved change template, evidence requirement, acceptance criteria and record-retention policy define authorised intent; current linked evidence and authorised operator statements show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Preserve actual observed and retrieved times, immutable evidence identifiers where available and the version of every test or state capture. Conflicts: Expose scope, implementation time, state, test result, backup, defect and owner conflicts; do not choose silently.

Answer boundary: Must not sign handback, close the change, alter evidence, expose restricted configuration or accept unresolved risk for the service owner.

Why this is useful: The team can leave the next shift and future investigators a usable record instead of an approval status with no traceable evidence of what changed or was tested.

7

Cybersecurity, identity, segmentation and access boundaries

IT-061 to IT-070 | sheet 1 of 5

IT-061

STAKEHOLDER QUESTION

Which segments support user, guest, building-service, management and infrastructure functions, and what authorised service boundaries apply between them?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: sanitised controller and routing policy state, VLAN or VRF identifiers, firewall policy objects, NAC service classes and bounded dependency tags across authorised network domains and their approved interconnection points. Coverage/timing: Capture daily snapshots plus event-driven policy and topology updates; summarise abstract segment-to-service boundary relationships without unnecessary addresses or rule detail. Quality/provenance: Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: overlays, shared services, temporary exceptions, NAT and supplier-managed controls can make the logical boundary different from simple VLAN membership. Do not expose credentials, full rule sets, management addresses, payloads or sensitive topology outside authorised cybersecurity and network roles.

Analysis required: Type: lookup. Decision interval: after design or policy change, before service onboarding and quarterly. Minimum evidence: a current approved service-flow matrix plus current sanitised state from every relevant enforcement layer. Method: model intended and observed boundaries separately, identify unobserved enforcement points and show only the level of detail permitted for the viewer. Uncertainty: report source age, clock quality and policy version, segment aliases, overlay networks and supplier-managed boundaries separately. Safe failure: return the boundary as partial and escalate to the policy owner when an enforcement layer or exception is unverified.

Authoritative supporting data: Owners: Cybersecurity, network architecture, network operations and relevant service owners. Precedence: the approved segmentation policy, service-flow matrix, exception records and architecture design define authorised intent; current sanitised controller, routing and policy state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat the observed boundary as stale after policy, routing, overlay or exception change and retain the effective version of the approved intent. Conflicts: Expose segment identity, service class, allowed flow, exception, management boundary and supplier-control conflicts; do not choose silently.

Answer boundary: Must not disclose sensitive topology or complete firewall policy broadly, authorise connectivity, change segmentation, grant access or certify isolation.

Why this is useful: The team can identify the correct segment and policy owner for a service, prevent accidental placement in the wrong trust zone and scope controlled connectivity testing.

IT-062

STAKEHOLDER QUESTION

Is any unexpected reachability present between segments that should be separated or limited to named authorised services?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: controlled synthetic probes between commissioned test endpoints, firewall allow or deny counters, route presence and approved flow-metadata summaries across specific source and destination segment pairs named in the authorised test plan. Coverage/timing: Capture on demand after policy changes and at a risk-based periodic interval, with low-rate test traffic; summarise allowed, denied, untested and path-unknown service pairs. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a successful probe tests only one protocol and path, asymmetric enforcement can mislead, and a failed probe may be caused by endpoint or service state. Do not perform broad scanning, inspect payloads or use real user devices and credentials.

Analysis required: Type: observation + comparison. Decision interval: after relevant changes, during authorised control validation and when an unexpected path is suspected. Minimum evidence: a current policy decision, healthy test endpoints, a permitted protocol and evidence from the relevant enforcement layer. Method: compare the bounded test with policy, distinguish endpoint failure from network denial and preserve untested services as unknown. Uncertainty: report source age, clock quality and test-point coverage, stateful controls, asymmetric paths and temporary policy exceptions separately. Safe failure: report unexpected reachability as a controlled-investigation finding and escalate without blocking or expanding the test.

Authoritative supporting data: Owners: Cybersecurity, network operations and the owners of the source and destination services. Precedence: the approved service-flow matrix, test authorisation, exception register and expected enforcement points define authorised intent; current controlled probe, route and enforcement-counter observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Bind results to the exact policy version, test endpoints and time; invalidate after route, firewall, endpoint or exception change. Conflicts: Expose flow intent, test scope, endpoint health, enforcement point, exception and observed result conflicts; do not choose silently.

Answer boundary: Must not scan unapproved ranges, exploit services, inspect content, block traffic, change firewall or routing policy, or certify that segmentation is complete.

Why this is useful: The team can verify a specific boundary, identify the enforcement layer that needs checking and avoid false assurance based only on configuration intent.

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Cybersecurity, identity, segmentation and access boundaries

IT-061 to IT-070 | sheet 2 of 5

IT-063

STAKEHOLDER QUESTION

Which privileged infrastructure identities and roles are current, justified, least-privileged and linked to an accountable owner?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: directory and privileged-access-management metadata for role, owner, approval, entitlement, last authorised use and credential-rotation state across privileged identities and groups for approved network, server, storage and management systems. Coverage/timing: Capture daily entitlement snapshots plus events for creation, role change, approval expiry and termination; summarise identity-to-role-to-system relationships and exception state. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: shared emergency accounts, nested groups, supplier identities, delayed HR or contract updates and incomplete target-system feeds can hide effective access. Never collect or display passwords, keys, tokens, session content or unnecessary personal information; reveal identity detail only to authorised identity-governance roles.

Analysis required: Type: lookup + comparison. Decision interval: daily for lifecycle events and at the authorised periodic access assessment. Minimum evidence: a verified identity or service identity, accountable owner, current approval, target-system entitlement and role definition. Method: expand nested memberships, separate standing from time-bound access, compare effective entitlement with approved role and flag ownerless or excess access. Uncertainty: report source age, clock quality and role inheritance, owner changes, emergency access and directory-to-platform mapping separately. Safe failure: return the entitlement as unverified and route to the identity and system owners rather than inferring legitimacy.

Authoritative supporting data: Owners: Identity governance, cybersecurity, privileged-access owners, system owners and accountable managers. Precedence: approved role definitions, access approvals, joiner-mover-leaver records and emergency-access procedures define authorised intent; current directory, PAM and target-system entitlement metadata show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply approval and employment or contract effective times; mark identities stale when the owner, role or target-system evidence is outdated. Conflicts: Expose identity, owner, role, approval, nested-group, target-system and emergency-access conflicts; do not choose silently.

Answer boundary: Must not reveal credentials, grant or revoke access, identify wrongdoing, approve entitlement, expose personal data broadly or replace the authorised identity process.

Why this is useful: The team can route ownerless or over-broad privilege to the correct approver and prepare precise evidence without handling credentials or secrets.

IT-064

STAKEHOLDER QUESTION

Which service, device, supplier or automation accounts are dormant, orphaned, duplicated or past their approved end date?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: identity-store, PAM and target-system metadata for account type, owner role, creation and end date, last successful use and approved service mapping across non-personal and supplier identities used by authorised infrastructure. Coverage/timing: Capture daily snapshots with lifecycle and authentication-state events; summarise account lifecycle and ownership status without command or session content. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: infrequent scheduled jobs, disaster-only identities, rotated aliases and incomplete target logs can make legitimate accounts appear dormant. Do not expose secrets, authentication payloads or unnecessary personal identifiers.

Analysis required: Type: lookup + comparison. Decision interval: daily for expired accounts and monthly for dormant or orphaned candidates. Minimum evidence: a verified account, current owner and purpose record, approved end date and sufficient usage coverage for its expected schedule. Method: compare lifecycle state with target presence and expected use, distinguish duplicate aliases and rank by privilege and service exposure. Uncertainty: report source age, clock quality and account reuse, service migration, stale ownership and incomplete last-used evidence separately. Safe failure: label the account status uncertain and request owner confirmation rather than assuming it can be disabled.

Authoritative supporting data: Owners: Identity governance, system owners, service owners, supplier managers and cybersecurity. Precedence: approved account purpose, owner, end date, exception and lifecycle process define authorised intent; current directory, PAM and target-system usage metadata show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the account's expected-use pattern and approval period; last use alone must not override an active approved disaster or break-glass purpose. Conflicts: Expose account alias, owner, purpose, end date, expected-use schedule and target-system conflicts; do not choose silently.

Answer boundary: Must not disable accounts, reveal credentials, attribute use to a person, declare compromise, extend approval or bypass supplier and identity controls.

Why this is useful: The team can identify high-risk ownerless access, preserve legitimate emergency identities and send a bounded disablement or ownership task through the authorised process.

7

Cybersecurity, identity, segmentation and access boundaries

IT-061 to IT-070 | sheet 3 of 5

IT-065

STAKEHOLDER QUESTION

Which infrastructure certificates or trust relationships are near expiry, invalid, or no longer linked to the approved service owner?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: certificate inventory metadata for service ID, issuer, validity, key type, chain state, deployment point and owner mapping across authorised network, server, storage, management and machine-to-machine services. Coverage/timing: Capture daily discovery plus event updates for issuance, deployment and revocation status where permitted; summarise expiry horizon, chain health, deployment coverage and owner state without private key material. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: hidden appliances, short-lived certificates, alternate endpoints, clock error and incomplete discovery can create omissions or false expiry alarms.

Analysis required: Type: lookup + forecast. Decision interval: daily with escalating horizons defined by the PKI owner. Minimum evidence: a verified service and endpoint mapping, current certificate metadata, trusted time and accountable owner. Method: calculate expiry windows, identify chain or hostname-policy mismatch under approved rules, and distinguish issued from actually deployed state. Uncertainty: report source age, clock quality and chain discovery, clock errors, hidden trust stores and ownerless dependencies separately. Safe failure: return coverage or validity as unverified and escalate to the PKI or service owner when discovery or time is unreliable.

Authoritative supporting data: Owners: PKI owners, service owners, platform operations and cybersecurity. Precedence: the approved certificate inventory, ownership, trust policy, renewal procedure and exception records define authorised intent; current endpoint and certificate-management observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use validity, revocation and deployment times and treat discovery as stale after endpoint, load-balancer or certificate change. Conflicts: Expose service ID, endpoint, owner, certificate serial, trust chain, clock and deployment conflicts; do not choose silently.

Answer boundary: Must not display private keys, renew or deploy certificates, alter trust stores, certify security or expose sensitive endpoints outside authorised roles.

Why this is useful: The team can prevent avoidable certificate outages, locate ownerless trust dependencies and schedule renewal and validation before the change becomes urgent.

IT-066

STAKEHOLDER QUESTION

Which network access-control points are not enforcing the approved authentication, device-admission or fallback method for their role?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: switch and wireless authentication mode, NAC policy state, RADIUS service health, aggregate admission outcomes and controlled synthetic admission tests across approved wired and wireless access points, switch ports and service classes. Coverage/timing: Capture controller state every 5 minutes, aggregate outcomes hourly and controlled tests after change or fault; summarise intended and observed admission method by enforcement point without real credential data. Quality/provenance: Validate test-point location, target, timeout, retry logic, software version and a known-good control. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: open ports, low device volume, cached sessions, fallback behaviour and unsupported edge devices can hide ineffective enforcement. Use synthetic identities and aggregate outcomes; do not collect real credentials, usernames, device content or session payloads.

Analysis required: Type: observation + comparison. Decision interval: daily for state, after relevant change and during authorised security diagnosis. Minimum evidence: a current policy decision, verified enforcement point, healthy identity service and either controlled test or corroborating outcome evidence. Method: compare configured, negotiated and tested methods, distinguish identity-service failure from edge fallback and preserve untested states as unknown. Uncertainty: report source age, clock quality and fallback behaviour, identity-service outages, cached state and enforcement-point coverage separately. Safe failure: report enforcement as unverified and escalate without bypassing or changing access control.

Authoritative supporting data: Owners: Cybersecurity, identity owners, NAC owners and network operations. Precedence: the approved admission policy, fallback rule, port or WLAN role and exception record define authorised intent; current controller, switch, WLAN, RADIUS and synthetic-test state show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Bind the result to the exact policy and enforcement-point version and invalidate after port-role, firmware, RADIUS or exception changes. Conflicts: Expose port or WLAN role, authentication method, fallback state, exception, test identity and policy-version conflicts; do not choose silently.

Answer boundary: Must not reveal credentials, bypass NAC, grant access, reconfigure ports or WLANs, quarantine devices or certify control effectiveness.

Why this is useful: The team can identify a misapplied role, fallback condition or identity-service dependency and take a precisely scoped correction through authorised change control.

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Cybersecurity, identity, segmentation and access boundaries

IT-061 to IT-070 | sheet 4 of 5

IT-067

STAKEHOLDER QUESTION

Which unsupported or known-vulnerable infrastructure assets create the greatest service exposure after network position, dependencies and compensating controls are considered?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: approved vulnerability findings, verified asset and version inventory, sanitised network reachability, service dependencies, control state and incident history across authorised infrastructure assets and their approved management or service paths. **Coverage/timing:** Capture asset state daily, vulnerability feeds at their approved cadence and reachability after relevant policy changes; summarise exposure factors and evidence quality by asset without exploit content. **Quality/provenance:** Validate collector identity, permissions, scope, schema or parser version, object mapping and secret redaction, and hash each snapshot. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. **Limits:** scanner blind spots, false positives, backported fixes, temporary exceptions and incomplete topology can materially change exposure. Do not exploit systems, collect payloads, expose detailed vulnerabilities broadly or identify individual users.

Analysis required: Type: lookup + estimate. Decision interval: daily for critical new findings and weekly for the prioritised queue. Minimum evidence: verified asset and version, applicable finding, actual permitted reachability or isolation evidence, service criticality and current compensating controls. Method: rank by plausible exposure and consequence, not score alone, and test sensitivity to uncertain topology or applicability. Uncertainty: report source age, clock quality and scanner applicability, asset identity, compensating-control evidence and network-position drift separately. Safe failure: return the priority as provisional and request product or network validation when key context is missing.

Authoritative supporting data: Owners: Cybersecurity, asset and product owners, network operations and service owners. Precedence: the accepted vulnerability record, vendor or internal applicability decision, risk policy and compensating-control evidence define authorised intent; current verified asset, reachability, dependency and control observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply finding, exception and control effective times; mark assets with uncertain version or position separately rather than assigning precise risk. Conflicts: Expose asset identity, vulnerability applicability, support status, reachability, service criticality and control conflicts; do not choose silently.

Answer boundary: Must not exploit, patch, isolate or reboot assets, declare compromise, approve risk acceptance or expose sensitive vulnerability and topology detail broadly.

Why this is useful: The team can focus scarce maintenance windows on assets with real service exposure and avoid wasting effort on findings that are inapplicable or strongly contained.

IT-068

STAKEHOLDER QUESTION

Which supplier or remote-support connections are active, approved for the current window and technically limited to the intended systems?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: VPN, zero-trust, PAM and jump-host session metadata for approved window, sponsor role, target group, start and end state, plus bounded reachability evidence across authorised supplier and remote-support access paths. **Coverage/timing:** Capture event-driven session state with 1-minute health and approval-window checks; summarise active, expired, out-of-window, target-bounded and state-unknown sessions. **Quality/provenance:** Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. **Limits:** shared supplier accounts, nested groups, emergency extensions, split tunnels and target-system logging gaps can obscure effective reach. Never collect credentials, keystrokes, session content or payloads unless a separately authorised security process requires it.

Analysis required: Type: lookup + observation. Decision interval: continuously while access is active and at session close. Minimum evidence: a verified supplier identity, current approval, sponsor, allowed target scope and current session metadata. Method: compare active state with time and target boundaries, expand effective group access where permitted and flag unobserved enforcement points. Uncertainty: report source age, clock quality and session visibility, jump-host layering, supplier clocks and emergency-window records separately. Safe failure: escalate out-of-window or scope-uncertain access to the authorised security owner without terminating or expanding it.

Authoritative supporting data: Owners: Remote-access owners, cybersecurity, PAM owners, service sponsors and contract managers. Precedence: the approved access request, time window, sponsor, target scope, method and emergency procedure define authorised intent; current VPN, PAM, jump-host and target-access metadata show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use actual session and approval times and invalidate access scope after policy, group or target changes; expired approval never becomes valid through continued connection. Conflicts: Expose identity, sponsor, window, target group, session state and emergency-extension conflicts; do not choose silently.

Answer boundary: Must not grant, extend, terminate or monitor content of remote sessions, reveal credentials, infer wrongdoing or replace cybersecurity incident procedures.

Why this is useful: The team can confirm that legitimate support is correctly bounded and route an expired or over-broad session to the authorised owner before it becomes unmanaged access.

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Cybersecurity, identity, segmentation and access boundaries

IT-061 to IT-070 | sheet 5 of 5

IT-069

STAKEHOLDER QUESTION

Do critical infrastructure systems have adequate security-event logging, protected retention, source health and time synchronisation for authorised investigation?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: log-source heartbeat, event-count summaries, collector queue state, retention and integrity status, schema version and NTP or PTP offset across critical network, server, storage, identity and management systems. Coverage/timing: Capture heartbeat and queue state every 1-5 minutes, clock offset at least daily and retention checks at the approved interval; summarise coverage, gap rate, clock quality, protection and retention by source class.

Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a healthy heartbeat does not prove every event type is present; quiet systems, filter changes, queue loss and clock steps can mislead. Do not display event payloads, identities or sensitive commands unless the authorised investigation role and purpose require them.

Analysis required: Type: observation + completeness check. Decision interval: daily for source loss, weekly for coverage and before relying on evidence in an incident. Minimum evidence: current source inventory, expected event classes, healthy collection path, retention evidence and clock offset within the authorised tolerance. Method: separate source silence from collector loss, identify unprotected or expired intervals and propagate clock uncertainty into timeline suitability. Uncertainty: report source age, clock quality and source coverage, protected retention, parser drift, queue loss and clock accuracy separately. Safe failure: state that evidence is incomplete for the affected interval and escalate rather than certifying forensic readiness.

Authoritative supporting data: Owners: Security logging owners, cybersecurity, system owners and records-governance owners. Precedence: the approved logging profile, retention, integrity, access and time-synchronisation requirements define authorised intent; current collector, source, storage and time-service observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Retain source and collector observed times, schema and filter versions; mark coverage stale after agent, filter, collector or clock changes. Conflicts: Expose source identity, expected event class, filter, retention, integrity and clock conflicts; do not choose silently.

Answer boundary: Must not expose log content broadly, alter retention or filters, certify legal or forensic sufficiency, identify a person as responsible or bypass access controls.

Why this is useful: The team can correct blind sources, queue or clock problems before an incident and know which time periods cannot support a reliable technical timeline.

IT-070

STAKEHOLDER QUESTION

Are aggregate scanning, connection-burst or cross-segment patterns sufficient to warrant authorised investigation without inspecting user content?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: NetFlow or IPFIX summaries, connection and denial counts, protocol and segment class, approved anomaly indicators and sensor health across authorised infrastructure and inter-segment observation points. Coverage/timing: Capture 1-5 minute flow aggregates with event alerts and time-of-week baselines; summarise pseudonymous source groups, destination service class, rate, fan-out and boundary crossing. Quality/provenance: Validate source identity, event schema, logging level, drop or queue counters, retention and collector coverage. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: software deployment, monitoring, vulnerability assessment, backup, discovery and failover can create legitimate bursts; NAT and sampling reduce attribution. Collect no packet payloads, user content or browsing history and minimise or protect endpoint identifiers.

Analysis required: Type: observation + diagnosis support. Decision interval: near real time for high-confidence indicators and daily for lower-confidence patterns. Minimum evidence: healthy observation point, enough baseline coverage for the same segment and time, and current authorised-activity context. Method: compare rate, fan-out and cross-segment pattern with robust baselines, exclude declared activity and report alternative operational explanations. Uncertainty: report source age, clock quality and baseline variability, legitimate management bursts, NAT aggregation and monitor placement separately. Safe failure: state only that authorised investigation may be warranted and preserve uncertainty when identity or sampling is inadequate.

Authoritative supporting data: Owners: Cybersecurity, network operations and owners of approved scanning or automation services. Precedence: the authorised activity schedule, segmentation policy, detection rules and investigation procedure define authorised intent; current aggregate flow, denial and sensor-health observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the rule and baseline version effective at the event time and mark sampled or missing intervals with their detection limits. Conflicts: Expose source class, approved activity, segment boundary, sampling rate, clock and event-definition conflicts; do not choose silently.

Answer boundary: Must not inspect payloads, identify a person, block or quarantine traffic, declare compromise, conduct active probing or replace cybersecurity triage.

Why this is useful: The team can give cybersecurity a narrow time, segment and pattern for investigation while avoiding intrusive collection and false certainty about benign operational bursts.

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Resilience, backup, growth, energy and lifecycle planning

IT-071 to IT-080 | sheet 1 of 5

IT-071

STAKEHOLDER QUESTION

Which critical services have current backups, and when was each complete recovery path last proven by a controlled restore?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: backup-job metadata, protected repository and immutability state, replication status, recovery-point age, integrity checks and authorised restore-test outcomes across critical infrastructure services, configurations, platform state and data classes within the approved backup scope. Coverage/timing: Capture job state at its scheduled cadence with daily summaries and test evidence at the approved interval; summarise coverage, freshness, protection and proven recovery by service and dependency. Quality/provenance: Validate job identity, protected scope, repository, retention, checksums and restore-test evidence. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: a successful job may omit dependencies, restore tests may cover only a sample, and key, licence, network or identity services can block recovery. Do not display backup contents, file names, credentials, keys or user data.

Analysis required: Type: lookup + observation. Decision interval: daily for failures, monthly for critical coverage and at continuity planning. Minimum evidence: verified service-to-backup mapping, current protected copy, integrity evidence, required dependencies and a valid restore test for the stated recovery scope. Method: separate backup success from recoverability, compare recovery-point age with objective and identify untested dependencies or steps. Uncertainty: report source age, clock quality and backup scope, restore-test age, dependency readiness and protected-secret availability separately. Safe failure: return "recovery path unproven" when the controlled test or a required dependency is missing.

Authoritative supporting data: Owners: Backup owners, service owners, platform or storage operations, cybersecurity and continuity owners. Precedence: the approved backup scope, retention, recovery procedure, recovery-point objective and restore-test requirement define authorised intent; current job, repository, replication, integrity and controlled test records show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat status as stale after service, platform, encryption, repository or dependency change and apply each service's required test age. Conflicts: Expose service scope, backup object, dependency, recovery point, test result, key custody and retention conflicts; do not choose silently.

Answer boundary: Must not retrieve or restore data, expose backup contents or keys, certify recoverability, alter retention or initiate continuity actions.

Why this is useful: The team can close real protection gaps, schedule the right restore test and avoid treating a successful backup job as proven recovery.

IT-072

STAKEHOLDER QUESTION

Where do single points of failure remain across network, compute, storage, identity, external service, power and cooling dependencies?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: current dependency graphs, device and cluster role state, path diversity, power and cooling mappings, supplier boundaries and recent controlled failure-test results across critical services and their authorised infrastructure dependencies. Coverage/timing: Capture topology and role state every 5-15 minutes, facilities state every 1-5 minutes and scenario evidence after controlled tests or changes; summarise confirmed and assumed dependency edges with redundancy state. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: shared ducts, racks, control planes, credentials, licences, suppliers and manual procedures may be absent from automated topology. Restrict sensitive resilience topology and supplier detail by role; collect no payloads or user content.

Analysis required: Type: lookup + calculation. Decision interval: quarterly, before major change and after resilience-relevant incidents. Minimum evidence: a current service graph covering every critical domain or an explicit visibility boundary, plus current role and redundancy state. Method: identify cut vertices and cut sets for one credible failure at a time, distinguish design assumptions from tested resilience and rank by service consequence. Uncertainty: report source age, clock quality and hidden shared dependencies, common-cause failures, stale diagrams and untested redundancy separately. Safe failure: return the result as incomplete and list unmodelled domains rather than declaring no single point of failure.

Authoritative supporting data: Owners: Service owners, network, platform, storage, identity and facilities operations, continuity owners and suppliers for their domains. Precedence: approved resilience architecture, dependency records, single-line diagrams, contracts and test outcomes define authorised intent; current controller, platform, facilities and service-health observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate the model after topology, hosting, feed, cooling, supplier or identity changes until mappings are confirmed. Conflicts: Expose dependency edge, redundancy role, physical co-location, supplier boundary, control-plane and test-result conflicts; do not choose silently.

Answer boundary: Must not certify resilience, trigger tests or failover, disclose sensitive architecture broadly, authorise investment or replace competent electrical and continuity judgement.

Why this is useful: The team can prioritise the shared switch, feed, controller, licence or supplier dependency whose failure would bypass nominal redundancy across several services.

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Resilience, backup, growth, energy and lifecycle planning

IT-071 to IT-080 | sheet 2 of 5

IT-073

STAKEHOLDER QUESTION

Given the people, systems, access and dependencies available now, what recovery-time and recovery-point range is defensible for each critical service?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: backup recovery-point age, restore-test durations, failover timings, runbook step times, platform capacity, dependency readiness and required access-state metadata across critical services and their approved recovery paths. Coverage/timing: Capture current readiness daily with evidence from the latest controlled exercise or recovery event; summarise time and data-loss ranges for each recovery stage and dependency. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: exercise conditions may be easier than a real outage, parallel work and supplier delay vary, and unavailable staff or access can dominate technical timings. Do not expose credentials, personal availability beyond authorised duty roles, backup content or sensitive runbook detail broadly.

Analysis required: Type: estimate + scenario test. Decision interval: quarterly, after exercises, after material change and during continuity activation planning. Minimum evidence: current objectives, an end-to-end recovery path, measured or defensible duration for each critical stage and current dependency readiness. Method: sum sequential stages, model parallel work conservatively, compare recovery-point age and test sensitivity to supplier, access and capacity delays. Uncertainty: report source age, clock quality and staff availability, access delays, supplier response, data age and untested stages separately. Safe failure: state that the objective cannot be evidenced and identify the dominant untested stage rather than promising recovery.

Authoritative supporting data: Owners: Service and continuity owners, infrastructure teams, identity owners, facilities and suppliers. Precedence: approved RTO and RPO, recovery strategy, runbook, staffing model, access requirement and supplier commitment define authorised intent; current readiness plus latest controlled recovery evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply objective and runbook effective dates; invalidate timings after major architecture, dependency, staffing or supplier changes. Conflicts: Expose objective, recovery scope, step duration, dependency readiness, staffing role and supplier-time conflicts; do not choose silently.

Answer boundary: Must not promise objective attainment, activate recovery, grant access, restore data, commit suppliers or certify business continuity.

Why this is useful: Continuity owners can accept the supported recovery range, add people or dependencies, narrow the service commitment or schedule a controlled recovery exercise.

IT-074

STAKEHOLDER QUESTION

When will network, compute, memory, storage, rack, power or cooling capacity become limiting under plausible demand and project-growth scenarios?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: aggregate utilisation, allocation, peak and headroom metrics from network, platform, storage, rack, PDU and cooling sources across authorised infrastructure domains and capacity pools serving the building. Coverage/timing: Capture 1-5 minute operational metrics retained as daily peaks and percentiles over 90 days or longer, plus approved project-demand inputs; summarise capacity baselines and scenario demand by domain and time horizon. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: structural breaks, seasonal use, new projects, thin provisioning, hidden supplier limits and sparse environmental capacity evidence widen forecasts. Use aggregate technical demand only; do not infer individual behaviour, research content or productivity.

Analysis required: Type: forecast. Decision interval: monthly, at project gates and before procurement cycles. Minimum evidence: verified effective capacity, representative demand history, current reserve rules and at least two plausible demand scenarios. Method: forecast each domain separately with conservative ranges, detect the first limiting constraint and test sensitivity to major uncertain projects. Uncertainty: report source age, clock quality and project uncertainty, structural breaks, reserve rules and hidden supplier limits separately. Safe failure: report a planning range or no defensible forecast when history is too short or the architecture is changing.

Authoritative supporting data: Owners: Capacity owners, service owners, facilities, project owners and finance or planning governance. Precedence: approved effective capacity, reserve and resilience rules, committed projects and planning assumptions define authorised intent; current operational telemetry and authorised project-demand estimates show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Reset or segment forecasts after architecture, metric, capacity or workload changes and retain assumption dates and owners. Conflicts: Expose effective capacity, reserve, project demand, scenario start, growth rate and supplier-limit conflicts; do not choose silently.

Answer boundary: Must not guarantee demand, approve procurement, infer personal activity, certify cooling or power capacity or present a forecast as a commitment.

Why this is useful: The team can order long-lead capacity work in the right domain and avoid expanding servers when the first real constraint is power, cooling, storage or network resilience.

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Resilience, backup, growth, energy and lifecycle planning

IT-071 to IT-080 | sheet 3 of 5

IT-075

STAKEHOLDER QUESTION

How much energy does local infrastructure use, which loads are material, and which can be reduced without weakening service obligations?

Sensors and telemetry: Proposed read-only evidence: calibrated UPS and PDU metering, device or host power telemetry where supported, cooling-energy allocation where metered, and asset-to-outlet mappings across authorised communications and server infrastructure and its directly attributable support load. Coverage/timing: Capture 30-60 second power data aggregated to 5-minute, daily and monthly energy summaries; summarise energy by room, rack and approved equipment or service class with uncertainty. Quality/provenance: Verify meter class, CT, phase, branch or outlet mapping, scaling and vendor self-test or calibration status. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: meter hierarchy can double count, shared cooling is hard to allocate, device estimates may be inaccurate and unmapped outlets create residual load. Do not infer individual work patterns or inspect workload content; use aggregate technical classes only.

Analysis required: Type: calculation + comparison. Decision interval: monthly, seasonally and before energy-saving proposals. Minimum evidence: verified non-overlapping meters or an explicit allocation method, representative operating periods and current service constraints. Method: reconcile upstream and downstream energy, quantify unallocated residual, separate fixed, variable and potentially controllable loads, and report ranges. Uncertainty: report source age, clock quality and meter hierarchy, shared-cooling allocation, unmapped load and factor validity separately. Safe failure: return only the measured boundary and residual uncertainty when attribution is not defensible.

Authoritative supporting data: Owners: Facilities energy owners, infrastructure operations, asset owners and sustainability governance. Precedence: the approved meter hierarchy, calibration, tariff or carbon factors, asset mapping and service operating constraints define authorised intent; current meter and supported device-power observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Treat evidence as stale after meter, feed, asset or cooling-allocation changes and retain factor effective dates and measurement uncertainty. Conflicts: Expose meter hierarchy, calibration, asset mapping, shared-load allocation, factor and operating-constraint conflicts; do not choose silently.

Answer boundary: Must not certify billing or carbon reporting, infer staff behaviour, power down equipment, change operating modes or promise savings.

Why this is useful: The team can focus engineering effort on material controllable loads and avoid pursuing apparent savings created by double counting or uncertain allocation.

IT-076

STAKEHOLDER QUESTION

Which assets should be refreshed first because of support status, failure history, capacity risk, resilience impact, energy use or unavailable spares?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: verified asset age and version, support lifecycle, incident and hardware-alarm history, capacity and energy metrics, redundancy position and spare availability across authorised network, compute, storage, UPS and management infrastructure. Coverage/timing: Capture asset and metric refresh daily to monthly according to source, with current lifecycle and contract updates; summarise evidence-backed refresh factors and uncertainty by asset. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: old assets can be stable, new assets can be critical, incident coding varies and energy or failure data may be incomparable across product classes. Do not expose sensitive vulnerabilities, topology or contract detail outside authorised planning roles.

Analysis required: Type: recommendation. Decision interval: quarterly and at budget or architecture planning gates. Minimum evidence: verified asset identity, support position, service consequence, at least one condition or capacity indicator and estimated replacement constraints. Method: score transparent criteria, avoid double counting correlated risks, produce priority bands and show how uncertain factors change ordering. Uncertainty: report source age, clock quality and failure-rate uncertainty, support-data age, spare availability and service criticality separately. Safe failure: leave assets incomparable or in an evidence-needed band when core inputs are missing.

Authoritative supporting data: Owners: Asset owners, infrastructure operations, cybersecurity, facilities, procurement and finance governance. Precedence: approved lifecycle policy, support status, risk appetite, architecture strategy and replacement constraints define authorised intent; current asset, incident, capacity, energy and spare evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply current vendor support and contract dates, and mark missing failure, energy or dependency evidence rather than assigning a neutral score. Conflicts: Expose asset identity, support status, criticality, incident attribution, energy basis, spare and strategy conflicts; do not choose silently.

Answer boundary: Must not approve replacement, procurement or disposal, certify failure risk, expose sensitive product weakness broadly or override architecture governance.

Why this is useful: The team can build a transparent refresh case, identify assets needing measurement before funding and protect critical unsupported equipment before an avoidable failure.

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Resilience, backup, growth, energy and lifecycle planning

IT-071 to IT-080 | sheet 4 of 5

IT-077

STAKEHOLDER QUESTION

Are the critical spares, licences, subscriptions and vendor entitlements available for the credible failures the team is expected to recover?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: spare-asset inventory and condition, location and access status, licence or subscription entitlement metadata, contract support level and tested replacement compatibility across approved critical network, server, storage, power and management failure scenarios. Coverage/timing: Capture inventory daily or after movement, entitlement and contract updates on change, and compatibility checks at the approved interval; summarise scenario-to-spare, entitlement and support coverage without exposing licence keys. Quality/provenance: Validate schema, effective dates, owner roles, record version and integration scope. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: spares can be incompatible, batteries and optics can age in storage, licences can depend on portals or accounts, and supplier lead times vary. Do not display licence keys, credentials, contract-sensitive details or exact secure-storage locations to unauthorised users.

Analysis required: Type: lookup + comparison. Decision interval: monthly for critical assets, before high-risk maintenance and at continuity planning. Minimum evidence: a credible failure scenario, current installed version, verified compatible spare or entitlement, authorised access route and replacement procedure. Method: map each scenario to required resources, test compatibility and lead-time assumptions, and distinguish on-site, supplier-held and unavailable coverage. Uncertainty: report source age, clock quality and part compatibility, licence scope, contract validity, storage location and lead times separately. Safe failure: state that recovery resource is unverified and escalate for controlled confirmation rather than counting an untested spare.

Authoritative supporting data: Owners: Asset and stores owners, product owners, procurement, contract managers and infrastructure operations. Precedence: the approved failure scenarios, spare policy, compatibility matrix, licence entitlement and support contract define authorised intent; current inventory, condition, entitlement and supplier evidence show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Apply expiry, warranty, subscription and storage-condition dates; mark a spare unproven after long storage or product and firmware changes. Conflicts: Expose part identity, compatibility, quantity, condition, secure location, entitlement and support-level conflicts; do not choose silently.

Answer boundary: Must not reveal keys or secure locations broadly, purchase or allocate stock, activate licences, promise supplier response or authorise replacement work.

Why this is useful: The team can fill the exact spare or entitlement gap that would block recovery and avoid discovering during an outage that the stocked item, licence or contract is unusable.

IT-078

STAKEHOLDER QUESTION

If the primary server room, communications room or core-network area is unavailable, which services have a verified alternative location, path and operating capacity?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: current alternative-path topology, standby platform and replication state, capacity and environmental telemetry, access readiness and controlled service-test evidence across critical services and their approved alternative locations or paths. Coverage/timing: Capture state every 1-5 minutes with readiness summaries daily and evidence from controlled exercises; summarise alternative-site or path readiness, capacity, dependency and test state. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: the alternative may share power, cooling, ducts, identity, licensing or supplier dependencies; access and as-built room status can change. Restrict continuity topology and secure-location detail by role; collect no application or backup content.

Analysis required: Type: estimate + scenario test. Decision interval: daily for critical readiness, quarterly for planning and during continuity activation decisions. Minimum evidence: verified alternative path or location, current capacity and environmental evidence, required access and dependencies, and a valid controlled test or explicit untested status. Method: evaluate one loss scenario, trace shared dependencies, compare surviving capacity with priority demand and show untested steps. Uncertainty: report source age, clock quality and alternative-site readiness, path diversity, capacity headroom and access constraints separately. Safe failure: state that alternative operation is unverified and escalate rather than assuming documented standby equals usable recovery.

Authoritative supporting data: Owners: Continuity and service owners, network and platform operations, facilities, identity owners and suppliers. Precedence: the approved alternative design, access plan, recovery procedure, capacity limits and dependency maps define authorised intent; current topology, standby, replication, facilities and controlled test observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Invalidate readiness after architecture, capacity, access, room, feed, cooling or supplier changes; the May 2021 floor plan cannot establish current alternative-room suitability. Conflicts: Expose location, access, shared dependency, capacity, replication, licence and exercise-result conflicts; do not choose silently.

Answer boundary: Must not activate the alternative, move workloads, grant access, certify room or electrical suitability, promise capacity or approve continuity action.

Why this is useful: The team can identify which services truly have an operational alternative and which require a test, capacity increase or dependency separation before the primary area is lost.

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Resilience, backup, growth, energy and lifecycle planning

IT-071 to IT-080 | sheet 5 of 5

IT-079

STAKEHOLDER QUESTION

Which consolidation, power-management, workload-placement or architecture options could reduce energy without violating service, capacity, resilience or security constraints?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: aggregate energy, utilisation, capacity, thermal, dependency, resilience and placement evidence plus current operating-state limits across authorised local infrastructure and feasible design options. Coverage/timing: Capture operational metrics over representative 30-90 day periods with scenario inputs refreshed at planning time; summarise baseline and option-level ranges for energy, capacity and risk. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: workload mobility, licensing, startup peaks, thermal effects, maintenance demand and standby requirements can invalidate apparent savings. Do not inspect workload content, individual activity or secrets; use aggregate service classes and approved constraints.

Analysis required: Type: comparison + recommendation. Decision interval: quarterly and at architecture or energy-planning gates. Minimum evidence: a reconciled energy baseline, current capacity and dependency evidence, explicit service constraints and at least two technically feasible options. Method: estimate ranges for energy and headroom, test credible component failure and peak demand, and reject options that rely on unverified mobility or redundancy. Uncertainty: report source age, clock quality and workload variability, rebound effects, migration risk and uncertain shared-energy allocation separately. Safe failure: return no defensible saving option when evidence or constraints are incomplete.

Authoritative supporting data: Owners: Infrastructure and service owners, cybersecurity, facilities energy and cooling owners, architecture and change governance. Precedence: approved service requirements, resilience and security constraints, licence terms and architecture standards define authorised intent; current energy, utilisation, placement, dependency and thermal observations show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Retain scenario assumptions and effective dates and mark options stale after architecture, workload or policy changes. Conflicts: Expose baseline, workload mobility, capacity, resilience, security, licence and thermal-effect conflicts; do not choose silently.

Answer boundary: Must not migrate workloads, power down equipment, change architecture, certify savings, override service or security requirements or approve a change.

Why this is useful: The team can develop energy proposals that preserve explicit resilience and service boundaries and identify the evidence or test required before implementation.

IT-080

STAKEHOLDER QUESTION

Which infrastructure investments should be prioritised when service risk, growth, resilience, cybersecurity, energy, support status, cost and evidence quality are compared?

Sensors and telemetry: No new physical sensor is justified. Proposed read-only evidence: verified outputs from capacity, incident, dependency, vulnerability, energy, lifecycle, backup and continuity analyses across authorised infrastructure investment options and affected service portfolios. Coverage/timing: Capture inputs refreshed quarterly and at budget or strategy gates; summarise option-level costs, benefits, risks, dependencies, uncertainty and evidence gaps. Quality/provenance: Validate each contributing source and back-test estimates where history permits. Record clock alignment, source health, gaps or drift, and source-to-summary lineage. Limits: benefits can overlap, costs may be incomparable, future projects are uncertain and weak telemetry can make precise ranking misleading. Use aggregated planning evidence and protect sensitive architecture, vulnerability and contract information by role.

Analysis required: Type: recommendation + scenario comparison. Decision interval: quarterly and at formal investment gates. Minimum evidence: defined options, current costs, service consequences, dependencies, uncertainty and an authorised criterion set. Method: normalise only comparable evidence, avoid double counting, produce priority bands and trade-offs, and show which assumptions change ordering. Uncertainty: report source age, clock quality and incomparable options, overlapping benefits, stale costs and governance-weight sensitivity separately. Safe failure: leave options incomparable or evidence-limited when a responsible decision cannot be supported.

Authoritative supporting data: Owners: Service and domain owners, cybersecurity, facilities, architecture, finance, procurement and authorised governance bodies. Precedence: approved strategy, risk appetite, service priorities, budget assumptions and decision criteria define authorised intent; current operational evidence and owner-validated option assumptions show current observation; an authorised owner or competent site check resolves exceptions. Time/provenance: store effective, observed and retrieved times, source/version, query or transform lineage, evidence link and role permission. Use the strategy, costs and evidence effective at the decision gate; mark stale inputs, unsupported benefits and missing owner approval explicitly. Conflicts: Expose option scope, cost, benefit, dependency, risk, double-counting and objective-weight conflicts; do not choose silently.

Answer boundary: Must not approve funding, procure or dispose of equipment, certify benefits, hide uncertainty, expose sensitive architecture broadly or replace authorised governance.

Why this is useful: Authorised governance can compare priority bands, expose assumptions that change the ordering, fund evidence gaps and choose without false numerical precision.

Catalogue summary: 80 questions across eight decision domains

Operational state, evidence limits, authority and safe failure remain visible from asset discovery through lifecycle investment.

Category	Question IDs	Primary decisions supported	Typical evidence families	Safe failure
1. Asset inventory, topology and service ownership	IT-001 to IT-010	Locate assets and owners; reconcile rack, port, power and hosting mappings; bound supplier responsibility.	CMDB and asset records, controllers, LLDP, rack/patch records, PDU state, service catalogue.	Unknown asset, owner, location or path triggers controlled validation.
2. Availability, latency, capacity and performance	IT-011 to IT-020	Triage outages and degradation; quantify headroom; compare baselines, targets and post-change behaviour.	Heartbeats, synthetic probes, counters, host/cluster/storage metrics and service targets.	Missing coverage is indeterminate; association does not prove cause.
3. Wired, wireless and building-network coverage	IT-021 to IT-030	Validate outlets, Wi-Fi, roaming, dense-use service, PoE and operational-location connectivity.	Cable tests, commissioned surveys, controller aggregates, managed test clients and current spatial records.	Unmeasured or stale spaces stay unknown; historical plans are not current fact.
4. Server-room power, cooling and environmental conditions	IT-031 to IT-040	Manage UPS, thermal, cooling, leak, power-quality, feed balance and runtime decisions.	UPS/PDU telemetry, rack sensors, BMS state, leak sensors, power-quality meters and controlled surveys.	Lost telemetry means unknown protection or environment, not normal.
5. Alerts, incidents, dependencies and service restoration	IT-041 to IT-050	Rank alerts, bound impact, test hypotheses, reconstruct timelines, validate recovery and hand over safely.	Alert metadata, dependency graphs, aligned events, synthetic checks, restoration tests and supplier status.	Critical uncertainty stays visible; no automatic root-cause claim.
6. Changes, maintenance windows and configuration evidence	IT-051 to IT-060	Coordinate windows, compare intended and observed state, verify safeguards and complete handback evidence.	Change records, baselines, sanitised state, backups, tests, exceptions and monitor health.	Evidence gaps remain with authorised change control; no automatic remediation.
7. Cybersecurity, identity, segmentation and access boundaries	IT-061 to IT-070	Check segmentation, privileged access, certificates, NAC, exposure, remote support and logging readiness.	Policy intent, bounded tests, directory/PAM metadata, certificates, vulnerability context and flow aggregates.	No credentials, payloads, person attribution, blocking or security certification.
8. Resilience, backup, growth, energy and lifecycle planning	IT-071 to IT-080	Validate recovery, expose single points, forecast growth, quantify energy and prioritise lifecycle investment.	Backup/restore evidence, dependency tests, capacity, energy, support, spare and continuity data.	Forecasts remain scenarios; authorised governance decides.

Evidence, permissions and provenance rules

These rules apply to every question and prevent an apparently precise answer from outrunning its evidence or authority.

1

Verify deployment

A proposed sensor or integration becomes evidence only after installation, mapping, commissioning, permission and current health are verified.

2

Preserve source roles

Keep approved intent, live observation, authorised human confirmation and analytic inference separate.

3

Record three times

Retain effective, observed and retrieved times. A current retrieval can contain an old observation.

4

Keep provenance

Retain source ID, version, query or transformation version and an evidence link for authorised tracing.

5

Enforce least privilege

Show only the detail needed for the viewer role. Credentials, identities, contracts and sensitive topology remain restricted.

6

Minimise data

Prefer counters, health state, synthetic tests and aggregates. Do not collect payloads, user content or unnecessary personal data.

7

Make staleness explicit

Every source needs a cadence and expiry rule. Stale evidence remains visible as stale; it is not treated as current.

8

Surface conflicts

Show disagreeing sources, owners and times. Do not average or select a convenient answer silently.

9

Propagate quality

Calibration or validation, clock quality, monitor health, coverage, mapping confidence, drift and gaps flow into the answer.

10

Use minimum evidence

Each analysis states the least evidence required. Without it, return unknown or escalate.

11

Separate association and cause

Temporal or topological association supports a hypothesis, not a causal finding without competent evidence.

12

Keep control external

Answers may inform authorised work, but they do not grant access, execute commands, restart services or approve changes.

Answer states and escalation routes

Every response should end in an action state without being mistaken for an operator, approver or competent authority.

SUPPORTED

Meaning

Minimum evidence is present, current and permitted; sources agree within stated uncertainty.

Action

Proceed with the bounded decision under normal owner authority and controls.

QUALIFIED

Meaning

A bounded answer is possible, but conflict, staleness or incomplete coverage materially affects confidence.

Action

Target a test, site check or owner confirmation before higher-consequence action.

UNKNOWN

Meaning

Required mapping, permission, source health or minimum evidence is absent.

Action

Do not infer normality or absence; request the missing evidence through the established route.

ESCALATE

Meaning

The decision concerns safety, access, cybersecurity, approval, electrical work, supplier authority or active control.

Action

Hand the evidence to the authorised owner or competent professional; the service remains advisory.

Role-based escalation map

Network operations

Ports, routing, switching, Wi-Fi, monitoring and approved connectivity tests.

Platform and storage operations

Servers, virtualisation, containers, storage, backups, clusters and platform capacity.

Cybersecurity and identity owners

Segmentation, privileged access, remote support, exposure, logging and authorised investigation.

Facilities and competent professionals

Electrical systems, UPS, cooling, leaks, room environment, physical access and safety.

Service owners and incident/change control

Service impact, acceptance criteria, incident authority, maintenance approval and operational handback.

Suppliers and contract owners

Carrier, cloud, vendor-managed infrastructure, support entitlement and contractual demarcation.